Who Said What and When: Speaker-Attributed Simultaneous Meeting Translation with Voice-Prompted Speech

Anonymous ACL submission

Abstract

A meeting interpreter must say what each person said, attribute it to the right person, and keep up, also when several people talk at once. Simultaneous speech-to-speech systems serve one speaker, and speaker-aware pipelines attach one speaker to each recognised stretch of speech, so concurrent speakers collapse into one. We present MIST, in which a speech LLM reads acoustic tokens interleaved with tokens of a frozen streaming diarizer (the interleaving of G-STAR), writes speaker-first, time-stamped translation segments for every speaker of an overlapping window, learns when to wait, and hands each segment to a speech generator prompted with that speaker's voice. Here *simultaneous* refers to the attributed text, which is committed as the meeting runs; the speech is queued and its listener-side delay is measured. On AMI reference windows MIST beats the strongest non-oracle diarize-then-translate cascade by 6.52 (English→Chinese) and 2.17 (Chinese→English) speaker-attributed BLEU. Scored separately on overlapped speech, it leads a speaker-blind model relabelled by the same diarizer, which recovers most of the overall gain, by 3.58 and 2.82 BLEU, and translates both speakers of an overlapping pair 76.3% vs. 51.8% of the time; it still drops more short overlapped interjections than the cascade, which a short-turn fallback partly restores. Run online over all 16 AMI and 20 AliMeeting test meetings, its reads keep pace with the audio (real-time factor 0.86–0.98 from measured per-read computation), where LocalAgreement falls behind (1.73–1.98); its text lag is no longer on overlapped than on single-speaker speech, and with one channel per speaker the median listener lag is 7.4 s in English→Chinese. One interpreter channel cannot keep up with overlapping speech, and on real Mandarin meetings MIST only ties with the cascade.

1 Introduction

A meeting interpreter renders each contribution into the listener's language as it unfolds and makes clear whose contribution it is: a "yes" from the project manager means something different from a "yes" from the client. And people talk at once: over a third of our AMI test windows (Carletta et al., 2005) hold several speakers.

Simultaneous speech-to-speech translation (SimulS2ST) systems translate one speaker with low latency and can keep that speaker's voice (Seamless Communication et al., 2023; Labiausse et al., 2025; Cheng et al., 2025a; Deng et al., 2025). Multi-speaker translation has precedents: DiariST (Yang et al., 2024) streams speaker-attributed speech-to-text translation of overlapping conversations with token-level serialisation, and X-Translator (Zhao et al., 2026) is a real-time speaker-aware speech-to-speech cascade with online voice prompts. They leave two gaps (Table 1). DiariST outputs text without timestamps or speech; X-Translator, like any pipeline that recognises one stream and attaches one speaker to each recognised span, cannot translate two people who talk at the same time, a limitation its authors state. And no multi-speaker translation system reports latency on overlapping meetings: DiariST reports none, X-Translator measures it on single-speaker clips.

We present MIST (*Meeting Interpretation with Speaker Tracking*), in which one decoder writes, for every speaker of a window, speaker-attributed, time-stamped translation segments $\langle \mathrm{spk}_k \rangle \langle \mathrm{t}_a \rangle$ translation $\langle \mathrm{t}_b \rangle$, which a separate speech generator speaks in the voice of speaker k (Figure 1). "End-to-end" here means this joint decoding from audio to attributed text; the diarizer, the audio encoder and the speech generator are frozen. The decoder reads a frozen streaming diarizer, whose slots stay fixed for the

Work	concurrent speakers	time stamps	streaming	latency	latency on overlap	speech	own voice
STAC-ST (2023)	two-party	–	–	–	–	–	–
DiariST (2024)	✓	–	✓	not reported	–	–	–
X-Translator (2026)	one per span	segment	✓	single speaker	–	✓	✓
SimulS2ST systems[a]	–	–	✓	✓	–	✓	one speaker
MIST	✓	✓	✓	text, compute, listener	✓	✓	✓

Table 1: What prior multi-speaker and simultaneous translation work supports and reports. Concurrent speakers: two people talking at once are translated as two speakers. Latency on overlap: measured on overlapping multi-party meetings. [a] Seamless Communication et al. (2023); Labiausse et al. (2025); Cheng et al. (2025a); Deng et al. (2025). Details: Table 28.

whole meeting, as speaker tokens interleaved with its acoustic tokens (from G-STAR (Peng et al., 2026); §3.3), so two concurrent voices stay two speakers; it writes speaker first, because a segment's voice must be known before its first word is spoken; it reads every second and writes until a wait token trained from the timing of each translated chunk (§3.7); and each slot is voiced from the diarizer's own solo regions (§3.8).

Our contributions are this decoding format with commit training and causal voice prompts, and an evaluation that isolates what it buys: overlap-aware scores, whole-meeting online runs with queued reads, cascade and relabelling controls on the same diarizer, and pre-registered tests (§4). On AMI reference windows MIST beats the strongest non-oracle cascade by 6.52 / 2.17 speaker-attributed BLEU (En→Zh / Zh→En). A speaker-blind model relabelled by the same diarizer recovers most of that gain overall, but not where speakers overlap: there MIST leads it by 3.58 / 2.82 BLEU and recovers both speakers of an overlapping pair far more often (§5.2); it drops more short overlapped interjections than the cascade. Online over whole meetings, the learned commit policy keeps pace with the audio where LocalAgreement does not, and its text lag does not grow with overlap (§5.3). Real Mandarin meetings and one-channel playback remain open (§5.5, §5.5).

2 Related Work

Simultaneous speech translation. Wait-k (Ma et al., 2019) writes after a fixed delay; LocalAgreement (Liu et al., 2020; Polák et al., 2022; Macháček et al., 2023) commits the prefix two consecutive hypotheses agree on, at a full decode per read; learned policies emit a wait token (Koshkin et al., 2024). SimulS2ST systems keep one speaker's voice (Zhang et al., 2024; Seamless Communication et al., 2023; Labiausse et al., 2025; Cheng et al., 2025a; He et al., 2026; Deng et al., 2025). Pre-segmented evaluation overstates deployability (Papi et al., 2025).

Multi-speaker recognition and translation. Serialised output training transcribes overlapping speakers in order (Kanda et al., 2020, 2022, 2021); streaming diarizers keep speaker order over long recordings (Park et al., 2024; Medennikov et al., 2025), and diarization can condition a recogniser (Polok et al., 2026; Huo et al., 2026) or be interleaved with a speech LLM's audio tokens (G-STAR; Peng et al., 2026), which we adopt. STAC-ST translates two-party calls offline with turn tokens (Zuluaga-Gomez et al., 2023); DiariST (Yang et al., 2024) streams speaker-attributed translation of overlapping speech with token-level serialisation and introduces speaker-attributed BLEU; X-Translator (Zhao et al., 2026) binds each committed recognition span to the speaker with the largest temporal overlap and synthesises it with that speaker's prompt. Voice-cloned translation lowers listeners' workload in online meetings (Ma et al., 2026), a motivation, not a validation of our voices. We compare with these systems by capability only (Table 1).

3 Method

3.1 Task: who said what, and when

Formulation. Let $\boldsymbol{x} \in \mathbb{R}^L$ be a single-channel recording of T seconds ($L = 16{,}000\,T$ samples) in which an unknown number of speakers talk, sometimes at once. The system outputs

$$\mathcal{Y} = \big((k_n,\, a_n,\, b_n,\, \boldsymbol{y}^{(n)})\big)_{n=1}^{N_\mathcal{Y}}, \qquad (1)$$

where $N_\mathcal{Y}$ is the number of segments, $k_n \in \{1, \ldots, S\}$ the speaker of segment n among at most S speakers, $a_n < b_n$ the start and end (s) of the source speech it translates, and $\boldsymbol{y}^{(n)}$ its translation, a string of target tokens. Labels are arbitrary but one person keeps one label for the whole meeting. In simultaneous mode segments are produced while $\boldsymbol{x}$ is recorded; in speech output $\boldsymbol{y}^{(n)}$ is spoken in the voice of k_n.

3.2 The acoustic stream: what is said

Encoder. Translation comes from Qwen3-Omni, whose LLM (the "Thinker") already translates single-speaker speech (Xu et al., 2025). Its frozen

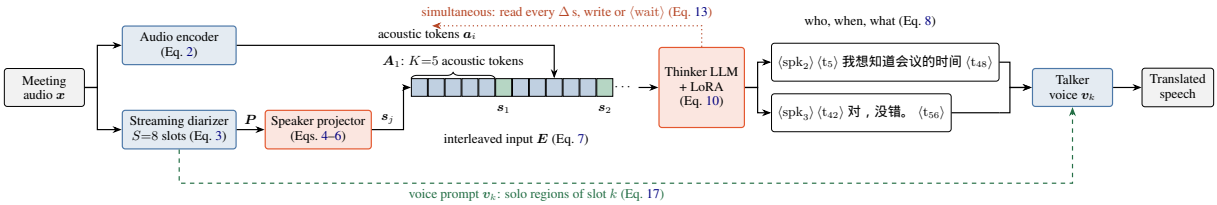


Figure 1: MIST. A frozen audio encoder turns the meeting into acoustic tokens (what is said); a frozen streaming diarizer gives slot activities $\boldsymbol{P}$ that a trained projector turns into speaker tokens $\boldsymbol{s}_j$ (who says it). One speaker token follows every block of K=5 acoustic tokens. The Thinker writes speaker-attributed, time-stamped translation segments, offline or read by read, and a Talker speaks each segment in the voice of its speaker slot. Blue: frozen; orange: trained. Gradients reach only the projector, the new token rows and the Thinker's LoRA adapters: who, what and when are decoded jointly, but the diarizer, the encoder and the Talker are not trained with them.

audio encoder maps the recording to acoustic tokens

$$\boldsymbol{A} = \mathrm{Enc}(\boldsymbol{x}) = [\boldsymbol{a}_1; \ldots; \boldsymbol{a}_N] \in \mathbb{R}^{N\times d}, \quad (2)$$

where row $\boldsymbol{a}_i \in \mathbb{R}^d$ describes about 80 ms of audio centred at $c_i = (i - \frac{1}{2})T/N$, so $N \approx 12.5\,T$, and $d = 2048$ is the Thinker's hidden size. They carry no memory of who spoke earlier in the meeting.

3.3 The speaker stream: who says it

Posteriors. A streaming diarizer keeps every voice in the slot it got when it first appeared, for the whole meeting (Medennikov et al., 2025). A frozen Streaming Sortformer, run in its streaming mode (chunks of 0.72 s, 0.32 s right context, a speaker cache of past frames) over a session, gives

$$\boldsymbol{P} = \mathrm{Diar}(\boldsymbol{x}) \in [0,1]^{F\times S}, \quad (3)$$

where $F = \lceil T/\delta \rceil$ frames of $\delta = 80$ ms, $S = 8$ slots, and $P_{f,k}$ is the probability that slot k talks in frame f; overlapped speech appears as several large entries in one row. To place speaker information next to the audio it describes, we resample $\boldsymbol{P}$ linearly in time to the token centres,

$$\tilde{\boldsymbol{P}} = \mathrm{Interp}(\boldsymbol{P};\, c_1, \ldots, c_N) \in [0,1]^{N\times S}, \quad (4)$$

so row $\tilde{\boldsymbol{p}}_i \in [0,1]^S$ is the slot activity at the centre of acoustic token i. A streaming frame is emitted once and never revised, so the frames a reader at time t uses (Eq. 12) are those a run on $\boldsymbol{x}_{\le t}$ alone gives (verified in Appendix K).

Projector. Activity has only $S = 8$ dimensions and changes slowly, so one speaker token per $K = 5$ acoustic tokens (400 ms) suffices. A strided convolution summarises each block of K rows ($\tilde{\boldsymbol{P}}$ zero-padded to MK rows, $M = \lceil N/K \rceil$),

$$\boldsymbol{z}_j = \sum_{r=1}^{K} \boldsymbol{W}^{(r)}\, \tilde{\boldsymbol{p}}_{(j-1)K+r} + \boldsymbol{b} \;\in \mathbb{R}^h, \quad (5)$$

for $j = 1, \ldots, M$, with kernels $\boldsymbol{W}^{(r)} \in \mathbb{R}^{h\times S}$, bias $\boldsymbol{b} \in \mathbb{R}^h$ and $h = 1024$. A two-layer perceptron lifts the summary into the Thinker's embedding space,

$$\boldsymbol{s}_j = \boldsymbol{W}_2\, \mathrm{GELU}\big(\boldsymbol{W}_1 \mathrm{GELU}(\boldsymbol{z}_j) + \boldsymbol{b}_1\big) + \boldsymbol{b}_2, \quad (6)$$

where $\boldsymbol{s}_j \in \mathbb{R}^d$ is the j-th speaker token, $\boldsymbol{W}_1 \in \mathbb{R}^{d\times h}$, $\boldsymbol{b}_1 \in \mathbb{R}^d$, $\boldsymbol{W}_2 \in \mathbb{R}^{d\times d}$ and $\boldsymbol{b}_2 \in \mathbb{R}^d$ (6.3M parameters in total). $\boldsymbol{W}_2$ and $\boldsymbol{b}_2$ start at zero, so training starts from the pretrained Thinker reading zero vectors and phases the speaker stream in only as far as it lowers the loss.

3.4 Interleaved fusion

Input. Following G-STAR (Peng et al., 2026), each speaker token follows the audio it describes. The audio part of the Thinker's input is

$$\boldsymbol{E} = [\boldsymbol{A}_1; \boldsymbol{s}_1;\, \boldsymbol{A}_2; \boldsymbol{s}_2;\, \ldots;\, \boldsymbol{A}_M; \boldsymbol{s}_M], \quad (7)$$

a matrix in $\mathbb{R}^{(N+M)\times d}$, where $\boldsymbol{A}_j = [\boldsymbol{a}_{(j-1)K+1}; \ldots; \boldsymbol{a}_{\min(jK,N)}]$ is the j-th block of at most K rows of Eq. (2) and $\boldsymbol{s}_j$ (Eq. 6) summarises the same block. $\boldsymbol{E}$ fills the audio placeholder of the chat template between an instruction $\boldsymbol{c}$ (Appendix B) and the answer; positions are consecutive.

3.5 Speaker-attributed, time-stamped output

Serialisation. We add 251 time tokens $\langle \mathrm{t}_0 \rangle, \ldots, \langle \mathrm{t}_{250} \rangle$ (one per 80 ms, up to 20 s) and $S = 8$ speaker tokens $\langle \mathrm{spk}_1 \rangle, \ldots, \langle \mathrm{spk}_8 \rangle$ in unused vocabulary rows, each with a trainable input embedding $\boldsymbol{e}_v \in \mathbb{R}^d$ and output row $\boldsymbol{o}_v \in \mathbb{R}^d$. A window's segments $\mathcal{Y}$ are written as

$$\boldsymbol{y} = \Big\Vert_{n=1}^{N_{\mathcal{Y}}} \big[\langle \mathrm{spk}_{k_n} \rangle\, \langle \mathrm{t}_{\alpha_n} \rangle\, \boldsymbol{y}^{(n)}\, \langle \mathrm{t}_{\beta_n} \rangle\big], \quad (8)$$

where $\|$ concatenates token sequences, segments are ordered by start as in serialized output training (Kanda et al., 2020), and $\alpha_n = \lfloor a_n/\delta \rceil$, $\beta_n = \lfloor b_n/\delta \rceil$ are start and end (from the window start) rounded to 80 ms steps and clipped to $[0, 250]$. Hesitations produce no segment. Rounding can make $\alpha_n = \beta_n$ for very short segments (2.6% of MIST's output segments in En→Zh); such a segment keeps its text but counts as zero-length speech in DER.

Slot-consistent targets. The reference names speakers, but the model sees diarizer slots, so targets must use slots. Per meeting we map each reference speaker $\rho \in \mathcal{R}$ to the slot that overlaps most with that speaker's speech over the *whole meeting*,

$$\sigma^\star = \arg\max_{\sigma} \sum_{\rho \in \mathcal{R}} \sum_{f=1}^{F_\mathrm{m}} G_{f,\rho} \mathbb{1}\big[P_{f,\sigma(\rho)} > 0.5\big], \tag{9}$$

where σ ranges over injective maps from reference speakers to slots, F_m is the number of frames of the meeting, and $G_{f,\rho} \in \{0, 1\}$ marks reference speech of ρ in frame f; the Hungarian algorithm (Kuhn, 1955) solves it. An utterance by ρ is labelled $k = \sigma^\star(\rho)$, so the model learns to copy the diarizer's slot rather than invent a label per window.

3.6 Training

Objective. Given $\boldsymbol{c}$ and $\boldsymbol{E}$, the Thinker with parameters θ defines

$$p_\theta(\boldsymbol{y} \mid \boldsymbol{x}) = \prod_{u=1}^{|\boldsymbol{y}|} p_\theta\big(y_u \mid \boldsymbol{y}_{<u}, \boldsymbol{E}, \boldsymbol{c}\big), \tag{10}$$

over the tokens y_u of Eq. (8). A wrong speaker or time is a single token that can misattribute a whole sentence, so we minimise the weighted cross-entropy

$$\mathcal{L} = -\frac{\sum_u \omega(y_u) \log p_\theta(y_u \mid \boldsymbol{y}_{<u}, \boldsymbol{E}, \boldsymbol{c})}{\sum_u \omega(y_u)}, \tag{11}$$

with $\omega = 2$ for speaker tokens, 1.5 for time tokens and 1 otherwise. The trainable parameters are the projector, the 2×259 new rows $\boldsymbol{e}_v, \boldsymbol{o}_v$, and LoRA adapters (Hu et al., 2022) ($r = 32$, $\alpha = 64$) on the query, key, value and output projections of all 48 Thinker layers. To keep plain translation, 15% of the rows are single-speaker translation rows; to avoid trusting the diarizer blindly, $\boldsymbol{P}$ is perturbed with probability 0.2 (Gaussian noise $\sigma = 0.1$, shift of up to one frame). The result, MIST (offline), decodes a window at once.

3.7 Simultaneous interpretation

Reading. The model reads every Δ seconds, at $t_r = \min(r\Delta, T)$ for $r = 1, \ldots, R = \lceil T/\Delta \rceil$. It sees the audio $\boldsymbol{x}_{\le t_r}$ and a causal speaker stream: the low-latency diarizer finalises a frame only $\lambda = 1.04$ s later, so later frames are zeroed,

$$P^{(r)}_{f,k} = P_{f,k} \mathbb{1}\big[f\delta \le t_r - \lambda\big], \tag{12}$$

for every slot k and frame f (covering $((f-1)\delta, f\delta]$). $\boldsymbol{E}^{(r)}$ is built from $\boldsymbol{x}_{\le t_r}$ and $\boldsymbol{P}^{(r)}$ as in Eqs. (2)–(7).

Writing and waiting. With $\hat{\boldsymbol{y}}^{(<r)}$ everything committed before read r, the model continues

$$\hat{\boldsymbol{y}}^{(r)} = \mathrm{Greedy}_\theta\big(\cdot \mid \hat{\boldsymbol{y}}^{(<r)}, \boldsymbol{E}^{(r)}, \boldsymbol{c}\big) \text{ until } \eta_r, \tag{13}$$

where the stop token is $\eta_r = \langle\mathrm{wait}\rangle$ for $r < R$ and $\eta_R = \langle\mathrm{eos}\rangle$. Everything before the stop token is committed and never revised. References and the slot map of Eq. (9) are used only for training targets and scoring; §3.9 removes the reference window end.

Deadline targets. For every read we need the part of the translation an interpreter would have committed by then. Each reference translation is split at punctuation into chunks $c_1, \ldots, c_J$, and each chunk is timed by the source it covers. With $|c|$ the length of a chunk (characters or words), chunk j covers the fraction

$$\phi_j = \frac{\sum_{j' \le j} |c_{j'}|}{\sum_{j'=1}^{J} |c_{j'}|} \in (0, 1] \tag{14}$$

of its utterance and ends at $e_j = \mathrm{end}(w_{\lceil \phi_j |\boldsymbol{w}| \rceil})$, where $\boldsymbol{w} = (w_1, \ldots, w_{|\boldsymbol{w}|})$ are the utterance's time-aligned source words. Where no word timings exist (the re-voiced Chinese source, §4), the chunk ends at the same fraction of the utterance, $e_j = a + \phi_j(b - a)$. A chunk becomes due one interpreting lag $\epsilon = 1$ s after its source, at the read

$$r(j) = \min\big(\{r : t_r \ge e_j + \epsilon\} \cup \{R\}\big). \tag{15}$$

The target of read r is the chunks due at r, each serialised as in Eq. (8), then the stop token,

$$\boldsymbol{y}^{(r)} = \Big(\mathop{\|}_{j:\, r(j)=r} \big[\langle \mathrm{spk}_{k_j} \rangle \langle \mathrm{t}_{\alpha_j} \rangle \, c_j \, \langle \mathrm{t}_{\beta_j} \rangle\big] \Big) \,\|\, \eta_r, \tag{16}$$

where k_j is the slot of the chunk's speaker under $\sigma^\star$ and α_j, β_j the chunk's own start and end. Each

read becomes one training row whose prefix is the targets of all earlier reads. MIST-Simul starts from the merged offline model and trains new LoRA adapters, the projector and the new rows on these rows with Eq. (11), one model per direction, at $\Delta = 1$ s (mixing details: Appendix D).

3.8 Speaking in each speaker's voice

Voices and layout. Slot k's solo frames are

$$m_{f,k} = \mathbb{1}\big[P_{f,k} > 0.5\big] \prod_{k' \neq k} \mathbb{1}\big[P_{f,k'} < 0.5\big], \tag{17}$$

where $m_{f,k} \in \{0, 1\}$; runs of solo frames of at least 0.4 s are concatenated into a voice prompt $\boldsymbol{v}_k$ of at most 8 s (a slot that never speaks alone uses its own segments). Offline we cut prompts from the whole session; in simultaneous mode only solo frames heard before the slot's first commitment are used, so no future audio is needed. The UniSS Talker (Cheng et al., 2025b), which predicts BiCodec tokens (Wang et al., 2025), speaks each slot's segments with prompt $\boldsymbol{v}_k$, giving segment n a duration D_n. One interpreter channel cannot play two segments at once, so segment n starts at

$$\pi_n = \max\big(\zeta_n,\ \pi_{n-1} + D_{n-1} + g\,\mathbb{1}[k_n \neq k_{n-1}]\big), \tag{18}$$

where $g = 0.2$ s separates speakers and ζ_n is the earliest start: the source start a_n offline (dubbing), or the commit time in simultaneous mode.

3.9 Running on a whole meeting

Online windows. A window opens at the first frame where any slot of the low-latency diarizer is active ($P_{f,k} > 0.5$) and closes after 1 s with no active slot or at 20 s (audio padded by 0.2 s / 0.3 s); this rule also cut part of the training windows. Nothing is filtered: all audio is processed, and how much of it is translated correctly is measured separately (§4). A window's opening and end are known only λ after the frames that decide them, so commits are delayed to the opening, and those of the last read to the moment the end is known. Slots persist across windows; the decoder restarts at each window. Segments are laid out on the meeting clock by Eq. (18), so a backlog carries over between windows; the reads of a meeting are queued on one decoder stream at the measured read time (Appendix Q).

4 Experimental Setup

Data. *En→Zh*: AMI (Carletta et al., 2005) IHM-mix recordings (all headsets summed into one channel), official split; every utterance is translated by three MT systems, one translation is selected by reference-free quality estimation and kept if its synthesised speech passes an audio check (Appendix A). Test windows are utterance groups (overlapping or touching utterances, at most 20 s, all passing the check): 2,375 windows from 16 meetings, 35% with several speakers. They keep 45.6% of the test speech and 35.6% of its overlap (Table 25); the whole-meeting runs translate all of it. Training uses 21,581 such windows from 137 meetings. *Zh→En*: each AMI utterance's Chinese translation is re-spoken by the UniSS Talker with one voice prompt per speaker and laid at its original start; the human AMI transcripts are the references (Appendix I; the same 16 meetings, so not an independent corpus). *Real Mandarin*: the AliMeeting test set (Yu et al., 2022), 20 meetings of two to four people (10.8 h), headsets summed into one channel, English references machine-translated from the human transcripts by an unrelated MT model; 4,243 windows, never used for training or tuning.

Models. Audio encoder and Thinker are Qwen3-Omni-30B-A3B-Instruct (Xu et al., 2025); the diarizer is Streaming Sortformer (Medennikov et al., 2025) with an offline preset and a low-latency preset (1.04 s look-ahead, λ in Eq. 12). MIST (offline) is one bidirectional model (2,000 steps, 67,795 rows). MIST-Simul continues from it with a new LoRA, one model per direction, on deadline-target reads (3,000 steps, 8 H200 GPUs); MIST-Simul-M (En→Zh) draws the read interval per window from $\{0.5, 1, 2\}$ s. Training uses Megatron-LM (Shoeybi et al., 2019) (Appendix D); MIST-Simul was trained with 3 seeds in En→Zh and 2 in Zh→En, every other model once. The Talker is a UniSS model (Cheng et al., 2025b) fine-tuned for block-wise simultaneous speech and shared by all systems. Decoding is greedy.

Baselines. *Cascades* translate the diarizer's turns one by one with our speaker-blind window model (*same-data*), with the same recipe trained on the cascade's own turns (*turn-trained*), with 2 s of context, or on *reference* turns and speakers (oracle input); the *streaming cascade* runs LA2 on each turn of the low-latency diarizer. *Relabelling*

is defined in §4. *Simultaneous*: LocalAgreement-2 (Liu et al., 2020; Polák et al., 2022) (LA2), which at every read decodes to the end and commits the common prefix with the previous continuation, on MIST (offline) and on the prefix-trained MIST-Simul model, and *fixed lag*, which commits complete segments predicted to end at least L s before the read. Details: Appendix E.

Metrics. cpBLEU, the translation analogue of concatenated minimum-permutation WER (Watanabe et al., 2020), is our primary window metric: per meeting, output slots are matched one-to-one to reference speakers by chrF (Popović, 2015), and BLEU is computed over the matched per-window pairs, unmatched text paired with the empty string (Appendix C). $\text{cpBLEU}_{\text{sess}}$ uses one matching per meeting and so rewards labels that stay with one person; $\text{cpBLEU}_{\text{win}}$ matches per window; $\text{BLEU}_{\text{blind}}$ ignores speakers. On whole meetings we score **tcpBLEU**, the analogue of time-constrained tcpWER (von Neumann et al., 2023): each output segment joins the reference unit of its matched speaker that it overlaps within a 2 s collar (Appendix C). We also report DER, speaker-count accuracy, text LAAL (Papi et al., 2022) and, on whole meetings, the *listener-side* lag from the end of a source segment to the end of its synthesised speech. **Speech** (dubbed offline, block by block in simultaneous mode) is transcribed by Paraformer or Whisper-large-v3 and scored with the same metrics, voices with two speaker-verification models (WavLM, TitaNet (Koluguri et al., 2022)). **Statistics**: exact sign-flip permutation tests over meetings and 95% meeting-bootstrap intervals in which every resample adds one draw of the decoder's run-to-run noise (Appendix F); v7's eight and v8's twelve pre-registered tests are each Holm-corrected (Appendices P, V).

Overlap-aware evaluation. Following the overlap decomposition of tcpWER, each reference unit of tcpBLEU is *overlapped* when another speaker talks during at least 20% of it, else *single*; tcpBLEU-ov and tcpBLEU-1spk are scored over each part (an insertion counts where two speakers talk). *Concurrent-speaker recall* (CR) takes every pair of utterances of two speakers that overlap by at least 0.5 s and asks whether both received text attributed to their own speaker within 1 s; *collapse* counts pairs where the missing side is covered by the other side's output speaker. We also report recall of short ($\leq$0.5 s, 0.5–1 s) utterances and units translated twice (Appendix V). En→Zh whole meetings are scored a second time on *all* 6563 non-hesitation units, using the utterance translations the windows were built from. *Post-hoc relabelling* gives each segment of the speaker-blind model the diarizer slot most active over its span, a one-speaker-per-span rule as in recognition-then-attribution pipelines; online it uses only the diarizer frames available at the segment's commit (*causal relabelling*). *Latency with queued reads*: a meeting is decoded by one decoder stream, so a read starts only when the previous one has finished, at the measured single-stream read time; RTF is read computation per second of meeting.

5 Results

A difference is called real only if an exact permutation test over meetings gives $p < 0.05$ (Holm-adjusted over the pre-registered tests, Tables 13 and 30) *and* its 95% interval with decoder noise excludes zero (Appendix F); otherwise we report a tie.

5.1 Offline: joint decoding against four cascades

MIST (offline) reaches 42.81 / 25.86 $\text{cpBLEU}_{\text{sess}}$ (En→Zh / Zh→En; Table 2); the cascade whose translator was trained on our windows reaches 36.29 / 22.36, and retraining it on its own diarizer turns helps only in Zh→En (35.27 / 23.69; MIST ahead by 7.54 and 2.17, p_{Holm} <0.001, <0.001). With reference turns and speakers (oracle input for the same translator, not an upper bound) it reaches 37.43 / 24.95. A speaker-blind model relabelled after decoding with the same diarizer reaches 42.06 / 24.66, close to MIST overall; the next section shows where the two differ.

5.2 Overlapped speech

Table 3 splits the scores by overlap (§4). On single-speaker units relabelling matches MIST (46.04 vs. 45.35 in En→Zh, Δ -0.69). On overlapped units MIST leads by 3.58 (En→Zh, 95% interval [2.56, 4.62]) and 2.82 (Zh→En, [2.32, 3.32]), and ties on AliMeeting (0.06). The gap opens with the amount of overlap (Figure 2): 46.11 vs. 46.88 on units nobody else talks over, 32.63 vs. 28.28 on units overlapped for more than 30% of their time. Relabelling gives one speaker per segment, so when two people talk at once one of them disappears

System	En→Zh LAAL s	En→Zh cpBLEU$_{sess}$	En→Zh BLEU$_{blind}$	En→Zh Spk	En→Zh DER	Zh→En LAAL s	Zh→En cpBLEU$_{sess}$	Zh→En BLEU$_{blind}$	Zh→En Spk	Zh→En DER
Offline										
Cascade, same-data translator	–	36.29	37.01	88.2	25.8	–	22.36	24.85	83.6	15.5
Cascade, turn-trained translator	–	35.27	35.27	79.3	37.8	–	23.69	25.32	86.4	16.9
Cascade, turns + 2 s context	–	23.75	25.02	88.3	25.9	–	16.88	19.53	83.5	15.6
MIST w/o speaker stream	–	21.17	43.40	84.8	54.5	–	12.20	28.13	84.9	57.4
+ diarizer relabelling	–	42.06	43.40	82.2	13.9	–	24.66	28.13	82.0	15.7
MIST (offline)	–	42.81†	43.33	89.5	14.0	–	25.86†	28.43	91.3	16.0
Simultaneous (Δ=1 s)										
MIST (offline) + LA2	2.37	31.64	32.74	87.8	15.2	2.50	24.81	27.67	90.7	17.3
MIST-Simul model + LA2	3.07	39.21	39.64	88.4	12.0	2.97	24.86	27.21	91.7	13.2
Streaming cascade (turns + LA2)	2.50	35.16	35.84	87.4	26.1	2.70	21.55	23.68	80.3	16.2
MIST-Simul	2.40	43.08±0.45^{3}	43.24	90.4	11.3	3.04	23.55±1.07^{2}	25.88	93.1	12.3
Oracle input (reference turns or activity), same translator										
Cascade, reference turns (oracle)	–	37.43	38.32	99.7	0.5	–	24.95	26.28	100.0	1.0
MIST (offline), oracle stream	–	42.32	42.95	92.2	14.2	–	27.32	28.37	95.9	11.6

Table 2: Speaker-attributed translation (text) on the reference windows of the AMI test meetings (2,375 En→Zh and 3,317 Zh→En windows, 16 meetings). One bidirectional MIST (offline) serves both directions; MIST-Simul is trained per direction and reported as the mean ± s.d. over its training seeds (superscript: number of seeds; per seed in Table 9). cpBLEU$_{sess}$ (primary) matches output speakers to reference speakers once per meeting; BLEU$_{blind}$ ignores speakers. Spk: windows with the right number of speakers (%); DER (%) of the output segments. LAAL includes the diarizer's 1.04 s look-ahead and is computation-unaware. †: better than the strongest non-oracle baseline of its block (exact permutation test over meetings $p < 0.05$ and the decoder-noise-widened interval excludes 0; Table 14).

System	En→Zh all	En→Zh overlap	En→Zh CR	Zh→En all	Zh→En overlap	Zh→En CR	AliMeeting all	AliMeeting overlap	AliMeeting CR
Reference windows, offline									
Cascade, same data	36.13	26.21	89.3	21.96	14.36	87.4	15.70	9.79	93.3
Speaker-blind + relabel	41.39	30.98	51.8	24.36	16.90	50.6	16.27	12.04	58.0
MIST (offline)	42.06	34.56	76.3	25.43	19.72	79.0	15.19	12.10	79.2
+ short-turn fallback	42.53	34.67	86.2	25.74	20.07	84.2	15.44	12.64	86.7
Whole meetings, online, 1 s reads									
Offline model + LA2	34.30	22.34	73.4	23.94	18.88	78.7	11.11	7.19	75.6
Simul model + LA2	39.31	32.91	75.5	23.96	17.89	79.4	11.91	7.60	76.7
Speaker-blind Simul + causal relabel	35.22	27.17	65.5	19.33	9.06	54.4	12.67	8.55	63.0
MIST-Simul	40.09	35.71	82.3	24.53	17.73	80.2	11.47	9.18	80.2

Table 3: Overlap-aware evaluation (§4): tcpBLEU over all units and over overlapped units, and concurrent-speaker recall (CR, %). Relabelling attaches one diarizer speaker to each translated segment. All rows and metrics: Table 31.

or is merged into the other (concurrent-speaker recall 51.8%, collapse 40.9%); MIST, reading both speakers' activity, writes both (76.3%, 20.3%; recall gain 0.245 / 0.284 / 0.211, all $p < 0.001$). This is the measurable contribution of reading the speaker stream inside the decoder. Against the cascade MIST is far ahead on overlapped units (8.36 / 5.36 / 2.32), but the cascade, which translates every diarizer turn on its own, recovers more concurrent pairs (89.3%) and more short overlapped utterances (67.8 vs. 60.0% of those under 0.5 s): MIST drops short interjections. Adding the cascade's translation of short turns MIST left out, a fixed rule, raises recall to 86.2% and tcpBLEU to 42.53 (Appendix V). Online, a speaker-blind Simul model trained identically and relabelled causally (pre-registered E-ov1/E-ov2, Table 30) trails MIST-Simul on overlapped units by 8.54 (En→Zh) and 8.67 (Zh→En) tcpBLEU, but ties on AliMeeting (0.63), where it is even ahead overall; concurrent-speaker recall is higher for MIST-Simul in all three sets (by 0.168 / 0.258 / 0.172).

5.3 Commit policy and latency

On reference windows LocalAgreement reaches 31.64 cpBLEU$_{sess}$ (En→Zh) on the offline model and 39.21 on the prefix-trained one; MIST-Simul's wait policy (42.63) is ahead of the latter within noise and 0.75 s earlier. On overlapped units the same comparison favours the learned policy

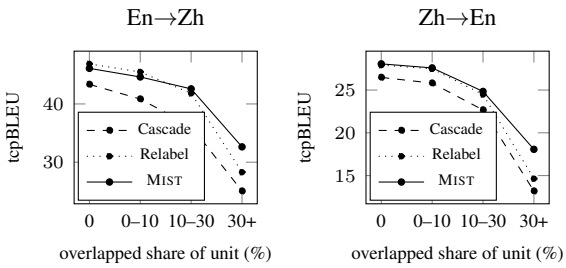


Figure 2: tcpBLEU by how much of a reference unit overlaps another speaker (reference windows, offline systems). The gap between MIST and post-hoc relabelling opens where speakers overlap.

			text lag p50 (s)		per-speaker channel (s)			1 ch.
Set	System	RTF	overlap	single	ovl. p50	single p50	p90	p50
En→Zh	MIST-Simul	0.86	1.59	1.96	6.8	7.7	12.2	142.4
	MIST + LA2	1.73	3.01	2.02	874.1	821.4	1688.6	903.0
Zh→En	MIST-Simul	0.72	1.73	1.95	7.5	8.5	15.1	26.4
	MIST + LA2	1.45	2.99	1.98	613.7	445.9	1009.6	500.1
AliMeeting	MIST-Simul	0.98	1.47	1.82	33.7	20.2	314.1	604.6
	MIST + LA2	1.98	4.09	1.99	1045.8	990.2	1783.9	1252.5

Table 4: Whole meetings with queued reads. RTF: read computation per second of meeting (>1: falls behind). Text lag: commit of a unit's last segment minus the unit's end. Per-speaker channel and one channel (1 ch.): completion lag of the speech heard.

(36.00 vs. 27.38; Table 31): LocalAgreement waits for two consecutive hypotheses to agree, which overlapping speakers rarely let happen. Its real cost is computation: LocalAgreement decodes to the end at every read (2028 ms per read, against 1008 ms). With the reads of a meeting queued on one decoder stream (Table 4) its real-time factor is 1.73–1.98, so it falls further behind the longer the meeting runs, whereas MIST-Simul's is 0.86 (En→Zh), 0.72 (Zh→En) and 0.98 (AliMeeting) and its median per-speaker listener lag rises only from 6.5 to 7.4 s in En→Zh when the reads are queued. Overlap does not slow it down: its text lag on overlapped units is no larger than on single-speaker ones (1.59 vs. 1.96 s; E-lat holds in all three sets, Table 30), and 54.2% of its output words are committed before the utterance they translate has ended. Across training seeds MIST-Simul's whole-meeting tcpBLEU is 40.14 ± 0.52 (En→Zh, 3 seeds); on the meeting subsets a fixed 2 s lag reaches the same or higher quality (behind on AMI overlapped units, ahead on AliMeeting) at about 1 s more lag on overlapped units (Appendix V).

5.4 Whole meetings, run online

We run the simultaneous systems over all 16 AMI and 20 AliMeeting test meetings with online windows (§3.9; Table 29): no reference boundary, all audio processed, commits delayed until a window's opening and end are known; the speaker stream is causal (Appendix K). Against LocalAgreement on the offline model (pre-registered E-on1–3) MIST-Simul is ahead by 5.83 in En→Zh (p_{Holm} <0.001) and ties in Zh→En (0.59) and on AliMeeting (0.38). Scored on all En→Zh units instead of the 45.6% the reference windows keep, the lead grows (39.24 vs. 27.01; overlapped units 33.56 vs. 16.61), so the filtered references did not flatter MIST. Against LocalAgreement on the same prefix-trained model (E-pol) its median text lag is 0.81 / 0.23 / 1.89 s shorter per meeting (Table 30) at tied quality (Table 3).

What a listener hears. One interpreter channel, which serialises overlapping speakers, cannot keep up: the median block of MIST-Simul is heard 142.4 s after its source ended in En→Zh and 604.6 s on AliMeeting. Captions for short overlapped interjections and skipping blocks more than 10 s late bound it (9.1 s) only by speaking just 80% of the units (the rest become captions) (Table 32). One channel per speaker keeps overlap as in the meeting: median 7.4 s (90th percentile 12.2 s), Zh→En 8.2 s, AliMeeting 23.2 s.

5.5 Real Mandarin meetings

On AliMeeting (Table 8) all scores drop. Offline MIST ties with the same-data cascade (14.90 vs. 15.25), beats the turn-trained cascade (E-cas3 1.85, p_{Holm} 0.001), and is behind relabelling overall (15.99) but not on overlapped units, where it still recovers far more concurrent pairs (79.2 vs. 58.0%). MIST-Simul and LocalAgreement tie on reference windows and whole meetings. The references are machine translations (Appendix T). In synthesised speech MIST leads the same-data cascade by less than in text (+1.63 / +1.97 cpASR-BLEU$_{\text{sess}}$; voices: Appendix O, untested by listeners).

6 Conclusion

MIST writes who, when and what for every speaker of an overlapping window as the meeting runs. Joint window translation carries its gain over cascades; reading the diarizer inside the decoder

pays off where people talk at once; its commit policy keeps pace with the audio, also under overlap. Short overlapped interjections, one-channel playback and real Mandarin meetings remain open.

Limitations

What was not done. No human evaluation: no listening test of voice identity or naturalness, and no bilingual check of the machine-translated En→Zh and AliMeeting references (by decision; AliMeeting references were scored only by an LLM judge, Appendix T). Not run for compute (prereg_v7 amendments 2–8, Appendix P): the ablations of the speaker-token rate K, the loss weights and the posterior noise; other fusions of the speaker stream; the deadline vs. stable-prefix control on matched data (Table 20 compares full-data deadline targets with stable-prefix targets on 8k windows); the diarizer-threshold sensitivity; decoding with a positive-length constraint; fixed lag in Zh→En and on all 16 En→Zh meetings; the streaming cascade over whole meetings; an uncontended wall-clock replay of whole meetings (the one contended replay completed is inconclusive, Appendix V), and a same-protocol comparison with DiariST (Yang et al., 2024) or X-Translator (Zhao et al., 2026) (compared by capability only, Table 1).

Evaluation scope. The reference windows keep 45.6% of the En→Zh test speech and 35.6% of its overlapped speech (Table 25); the whole-meeting runs read all speech but score the same reference utterances, so dropped speech can only cost insertions. The two AMI directions share their 16 meetings and are not independent replications; the Zh→En source is re-voiced, not real. All audio is close-talk microphones mixed into one channel; far-field audio is untested. Greedy bf16 decoding is not batch-invariant; decoders of different layouts and batch sizes enter only through the widened intervals, estimated from two re-decodes.

Baselines and training. The cascades, relabelling and LocalAgreement share our diarizer and translator; published systems were not re-run under our protocol (Table 1). MIST-Simul has 3 training seeds in En→Zh and 2 in Zh→En, whose seeds differ by more than a point; every other model was trained once. Reference speaker activity does not help En→Zh, so a better diarizer need not give better translations.

Latency and speech. Text LAAL is computation-unaware; the queued listener-side lag uses the median single-stream read time on H200 cards, not a wall-clock replay, and every read re-encodes its whole prefix without a key-value cache. One interpreter channel serialises overlapping speech and builds a backlog; captions and skipping bound the lag only by not speaking part of the output. Committed output is never revised, so a wrong speaker label is spoken. Slots are anonymous. Speech is scored by ASR and two speaker-verification models, and voice prompts come from the diarizer's solo regions, which a speaker who never talks alone lacks.

Ethics Statement

Speaker-attributed interpretation processes the voices of meeting participants and reproduces them in another language. Voice cloning can be misused to impersonate people; a deployed system should obtain the participants' consent and mark synthesised speech as such. AMI is distributed under a CC BY 4.0 licence and was recorded with the participants' consent; AliMeeting is used for testing only, under its licence.

References

Jean Carletta, Simone Ashby, Sebastien Bourban, Mike Flynn, Mael Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner. 2005. The AMI meeting corpus: A pre-announcement. In *Machine Learning for Multimodal Interaction (MLMI)*, pages 28–39.

Shanbo Cheng et al. 2025a. Seed LiveInterpret 2.0: End-to-end simultaneous speech-to-speech translation with your voice. *arXiv preprint arXiv:2507.17527*.

Sitong Cheng et al. 2025b. UniSS: Unified expressive speech-to-speech translation with your voice. *arXiv preprint arXiv:2509.21144*.

Keqi Deng, Wenxi Chen, Xie Chen, and Phil Woodland. 2025. SimulS2S-LLM: Unlocking simultaneous inference of speech LLMs for speech-to-speech translation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16718–16734, Vienna, Austria. Association for Computational Linguistics.

Zhifu Gao, Zerui Li, Jiaming Wang, Haoneng Luo, Xian Shi, Mengzhe Chen, Yabin Li, Lingyun Zuo, Zhihao Du, Zhangyu Xiao, and Shiliang Zhang.

2023. FunASR: A fundamental end-to-end speech recognition toolkit. In *Proc. Interspeech*.

Rongshen He, Xinyu Liang, et al. 2026. SimulS2ST-Omni: Data-efficient streaming speech-to-speech translation via explicit trajectory supervision. *arXiv preprint arXiv:2607.19810*.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.

Mingyue Huo, Yiwen Shao, and Yuheng Zhang. 2026. TagSpeech: End-to-end multi-speaker ASR and diarization with fine-grained temporal grounding. *arXiv preprint arXiv:2601.06896*.

Masoud Jalili Sabet, Philipp Dufter, François Yvon, and Hinrich Schütze. 2020. SimAlign: High quality word alignments without parallel training data using static and contextualized embeddings. In *Findings of EMNLP*.

Naoyuki Kanda, Yashesh Gaur, Xiaofei Wang, Zhong Meng, and Takuya Yoshioka. 2020. Serialized output training for end-to-end overlapped speech recognition. In *Proc. Interspeech*, pages 2797–2801.

Naoyuki Kanda, Jian Wu, Yu Wu, Xiong Xiao, Zhong Meng, Xiaofei Wang, Yashesh Gaur, Zhuo Chen, Jinyu Li, and Takuya Yoshioka. 2022. Streaming multi-talker ASR with token-level serialized output training. In *Proc. Interspeech*, pages 3774–3778.

Naoyuki Kanda, Guoli Ye, Yashesh Gaur, Xiaofei Wang, Zhong Meng, Zhuo Chen, and Takuya Yoshioka. 2021. End-to-end speaker-attributed ASR with transformer. In *Proc. Interspeech*, pages 4413–4417.

Tom Kocmi and Christian Federmann. 2023. Large language models are state-of-the-art evaluators of translation quality. In *Proceedings of the 24th Annual Conference of the European Association for Machine Translation*, pages 193–203, Tampere, Finland. European Association for Machine Translation.

Nithin Rao Koluguri, Taejin Park, and Boris Ginsburg. 2022. TitaNet: Neural model for speaker representation with 1D depth-wise separable convolutions and global context. In *Proc. ICASSP*.

Roman Koshkin, Katsuhito Sudoh, and Satoshi Nakamura. 2024. TransLLaMa: LLM-based simultaneous translation system. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 461–476.

Harold W. Kuhn. 1955. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97.

Tom Labiausse, Laurent Mazaré, Edouard Grave, Patrick Pérez, Alexandre Défossez, and Neil Zeghidour. 2025. High-fidelity simultaneous speech-to-speech translation. *arXiv preprint arXiv:2502.03382*.

Danni Liu, Gerasimos Spanakis, and Jan Niehues. 2020. Low-latency sequence-to-sequence speech recognition and translation by partial hypothesis selection. In *Proc. Interspeech*.

Mingbo Ma, Liang Huang, Hao Xiong, Renjie Zheng, Kaibo Liu, Baigong Zheng, Chuanqiang Zhang, Zhongjun He, Hairong Liu, Xing Li, Hua Wu, and Haifeng Wang. 2019. STACL: Simultaneous translation with implicit anticipation and controllable latency using prefix-to-prefix framework. In *Proceedings of ACL*.

Yong Ma, Yuchong Zhang, Peter Andrews, Zhikun Wu, Stephanie Zubicueta Portales, and Morten Fjeld. 2026. "Same Voice, Different Language": An exploration of voice-cloned translation to support non-native speakers in online meetings. In *Proceedings of the 31st International Conference on Intelligent User Interfaces*, pages 1742–1759. Association for Computing Machinery.

Dominik Macháček, Raj Dabre, and Ondřej Bojar. 2023. Turning Whisper into real-time transcription system. In *Proceedings of IJCNLP-AACL: System Demonstrations*.

Ivan Medennikov, Taejin Park, Weiqing Wang, He Huang, Kunal Dhawan, Jinhan Wang, Jagadeesh Balam, and Boris Ginsburg. 2025. Streaming Sortformer: Speaker cache-based online speaker diarization with arrival-time ordering. In *Proc. Interspeech*.

Sara Papi, Marco Gaido, Matteo Negri, and Marco Turchi. 2022. Over-generation cannot be rewarded: Length-adaptive average lagging for simultaneous speech translation. In *Proceedings of the Third Workshop on Automatic Simultaneous Translation*.

Sara Papi, Peter Polák, Dominik Macháček, and Ondřej Bojar. 2025. How "real" is your real-time simultaneous speech-to-text translation system? *Transactions of the Association for Computational Linguistics*, 13:281–313.

Taejin Park, Ivan Medennikov, Kunal Dhawan, Weiqing Wang, He Huang, Nithin Rao Koluguri, Krishna C. Puvvada, Jagadeesh Balam, and Boris Ginsburg. 2024. Sortformer: Seamless integration of speaker diarization and ASR by bridging timestamps and tokens. *arXiv preprint arXiv:2409.06656*.

Jing Peng, Ziyi Chen, Haoyu Li, Yucheng Wang, Duo Ma, Mengtian Li, Yunfan Du, Dezhu Xu, Kai Yu, and Shuai Wang. 2026. G-STAR: End-to-end global speaker-tracking attributed recognition. *arXiv preprint arXiv:2603.10468*.

Peter Polák, Ngoc-Quan Ngoc, Tuan-Nam Nguyen, Danni Liu, Carlos Mullov, Jan Niehues, Ondřej Bojar, and Alexander Waibel. 2022. CUNI-KIT system for simultaneous speech translation task at IWSLT 2022. In *Proceedings of the 19th International Conference on Spoken Language Translation (IWSLT)*.

Alexander Polok, Dominik Klement, Martin Kocour, Jiangyu Han, Federico Landini, Bolaji Yusuf, Matthew Wiesner, Sanjeev Khudanpur, Jan Černocký, and Lukáš Burget. 2026. DiCoW: Diarization-conditioned Whisper for target speaker automatic speech recognition. *Computer Speech & Language*, 95.

Maja Popović. 2015. chrF: Character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395.

Seamless Communication, Loïc Barrault, et al. 2023. Seamless: Multilingual expressive and streaming speech translation. *arXiv preprint arXiv:2312.05187*.

Mohammad Shoeybi, Mostofa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. 2019. Megatron-LM: Training multi-billion parameter language models using model parallelism. *arXiv preprint arXiv:1909.08053*.

Thilo von Neumann, Christoph Boeddeker, Marc Delcroix, and Reinhold Haeb-Umbach. 2023. MeetEval: A toolkit for computation of word error rates for meeting transcription systems. In *Proc. CHiME 2023 Workshop*.

Xinsheng Wang et al. 2025. Spark-TTS: An efficient LLM-based text-to-speech model with single-stream decoupled speech tokens. *arXiv preprint arXiv:2503.01710*.

Shinji Watanabe, Michael Mandel, Jon Barker, Emmanuel Vincent, Ashish Arora, Xuankai Chang, Sanjeev Khudanpur, Vimal Manohar, Daniel Povey, Desh Raj, et al. 2020. CHiME-6 challenge: Tackling multispeaker speech recognition for unsegmented recordings. In *Proc. CHiME Workshop*, pages 1–7.

Jin Xu et al. 2025. Qwen3-Omni technical report. *arXiv preprint arXiv:2509.17765*.

Mu Yang, Naoyuki Kanda, Xiaofei Wang, Junkun Chen, Peidong Wang, Jian Xue, Jinyu Li, and Takuya Yoshioka. 2024. DiariST: Streaming speech translation with speaker diarization. In *2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 10866–10870.

Fan Yu, Shiliang Zhang, Yihui Fu, Lei Xie, Siqi Zheng, Zhihao Du, Weilong Huang, Pengcheng Guo, Zhijie Yan, Bin Ma, Xin Xu, and Hui Bu. 2022. M2MeT: The ICASSP 2022 multi-channel multi-party meeting transcription challenge. In *Proc. ICASSP*, pages 6167–6171.

Split	Mtg.	Windows	Hours	Mean s	Utt.	Multi %	Ovl. %
En→Zh							
Train	137	21,581	36.8	6.1	54,824	38.5	38.2
Dev	18	3,203	5.2	5.9	8,206	40.0	39.6
Test	16	2,375	3.9	5.9	5,790	35.0	34.7
Zh→En							
Train	127	27,953	48.5	6.3	69,540	37.4	34.6
Dev	15	3,366	5.7	6.1	8,340	39.3	36.6
Test	16	3,317	5.8	6.3	8,634	36.5	34.5
Zh→En, AliMeeting (real meetings)							
Test	20	4,243	7.5	6.4	9,730	33.4	33.2

Table 5: Reference windows of the two directions. En→Zh: AMI IHM-mix with Chinese references; Zh→En: the re-voiced Chinese meetings (Appendix I) with the AMI English transcripts as references. *Multi*: windows with two or more speakers; *Ovl.*: windows in which two utterances overlap.

Shaolei Zhang, Qingkai Fang, Shoutao Guo, Zhengrui Ma, Min Zhang, and Yang Feng. 2024. StreamSpeech: Simultaneous speech-to-speech translation with multi-task learning. In *Proceedings of ACL*.

Yuxiang Zhao, Yichi Zhang, Yanjie An, Yanqiao Zhu, Zhanxun Liu, Yushen Chen, Qixi Zheng, Haina Zhu, Yunchong Xiao, Keqi Deng, Shuai Fan, Kai Yu, and Xie Chen. 2026. X-Translator: A real-time multilingual speaker-aware speech-to-speech translation system. *Preprint*, arXiv:2607.17544. ArXiv preprint, July 2026.

Juan Zuluaga-Gomez, Zhaocheng Huang, Xing Niu, Rohit Paturi, Sundararajan Srinivasan, Prashant Mathur, Brian Thompson, and Marcello Federico. 2023. End-to-end single-channel speaker-turn aware conversational speech translation. In *Proceedings of EMNLP*, pages 7255–7274.

A Data Construction

Translations. Every AMI utterance (IHM segmentation, official train/dev/test meetings) is translated by three MT systems (Hy-MT2-30B-A3B, NiuTrans LMT-60-4B and MiLMMT-46-4B). One candidate is selected with reference-free quality estimation (XCOMET-XL and an XLM-R-XL estimator) and the chrF agreement between candidates. The selected translation is also synthesised in the speaker's voice (VoxCPM2), and the synthesis is checked for audibility and intelligibility (Whisper large-v3 CER). Of the 12,643 test utterances, 96.3% received a translation and speech.

Review. A window enters our data only if all its non-hesitation utterances have a non-empty translation whose speech decodes and is not silent. A stricter review also requires the speech not to be over-long and to be recognised with CER ≤ 0.3; it rejects many windows with short back-channels and is used only as a test subset (1,454 of 2,375 test windows).

Windows. Reference windows are AMI utterance groups: utterances that overlap or touch are grouped, and a group longer than 20 s is dropped. Diarizer windows (training only) open at the first active diarizer frame and close after 1 s without activity or at 20 s; they are padded by 0.2 s before and 0.3 s after. A window is dropped if an utterance overlapping it has less than 90% of its duration inside the window.

B Instruction

The instruction $\boldsymbol{c}$ is the Thinker's speech translation instruction followed by: *"Several people are speaking. For every utterance write* <|spk k|><|t a|>*translation*<|t b|>*: k is the speaker's slot in the speaker stream, a and b are its start and end in 80 ms steps from the start of the audio, utterances in order of start."* The prompted baseline receives the same audio without speaker tokens and the plain translation instruction.

C Metrics

cpBLEU. Let meeting m consist of windows $\mathcal{W}_m$. For output slot k and window w, let $H_{w,k}$ be the concatenated text of all segments with label k, and for reference speaker ρ let $G_{w,\rho}$ be the concatenated reference translations of ρ in w. The session-level matching is

$$\pi_m = \arg\max_{\pi} \sum_{k} \mathrm{chrF}\big(H_{m,k},\, G_{m,\pi(k)}\big), \quad (19)$$

where $H_{m,k} = \|_{w\in\mathcal{W}_m} H_{w,k}$ and $G_{m,\rho} = \|_{w\in\mathcal{W}_m} G_{w,\rho}$ are the meeting-level concatenations and π ranges over one-to-one partial maps from output slots to reference speakers. $\text{cpBLEU}_{\text{sess}}$ is corpus BLEU over the pairs $(H_{w,k}, G_{w,\pi_m(k)})$ of all windows and matched slots, plus $(H_{w,k}, \varnothing)$ for unmatched output slots and $(\varnothing, G_{w,\rho})$ for unmatched reference speakers; missing or extra speakers are therefore penalised through brevity and precision. $\text{cpBLEU}_{\text{win}}$ uses a separate matching π_w for every window.

DER. Output segments (k_n, a_n, b_n) and reference utterances (hesitations excluded) are rasterised on 10 ms frames; output labels are mapped to reference speakers per meeting by maximal total overlap (Hungarian), and missed speech, false alarm and confusion are summed over all windows without a collar.

LAAL. For a window of duration T whose committed output has $|\hat{\boldsymbol{y}}|$ characters with commit times $d_1 \le \cdots \le d_{|\hat{\boldsymbol{y}}|}$ and whose reference has $|\boldsymbol{y}^\star|$ characters,

$$\mathrm{LAAL} = \frac{1}{\tau}\sum_{i=1}^{\tau}\Big(d_i - (i-1)\frac{T}{\max(|\hat{\boldsymbol{y}}|, |\boldsymbol{y}^\star|)}\Big), \quad (20)$$

where τ is the index of the first character committed at or after T (or $|\hat{\boldsymbol{y}}|$). We average LAAL over windows.

tcpBLEU. Whole meetings have no windows to pair output with reference, so we score time-constrained: (1) reference units are a speaker's scored utterances chained while less than 1 s apart (a model may write two consecutive utterances of one person as one segment); (2) output slots are matched to speakers per meeting by chrF over the whole meeting's text, as for cpBLEU; (3) each output segment, on the meeting clock and widened by a collar c on both sides, goes to the annotated utterance it overlaps most: if that utterance was not scored (a hesitation, or speech the reference windows dropped) the segment is ignored; if it belongs to the segment's matched speaker the segment joins that utterance's unit; otherwise it joins the matched speaker's own unit that it overlaps most, and if there is none it is an insertion. A segment spanning two units counts once, for the one it overlaps most; several segments in one unit are joined in time order; a segment overlapping no annotated speech is an insertion. tcpBLEU is corpus BLEU over (unit hypothesis, unit reference) pairs plus (insertion, empty) pairs. *Example*: speakers A and B, A says u_1 (0–4 s) and u_2 (4.5–8 s), B says "yes" (2–2.5 s) inside u_1 and u_3 (6–9 s) overlapping u_2. A's units: $\{u_1 u_2\}$ (0.5 s apart); B's: $\{\text{yes}\}$, $\{u_3\}$. An output segment of A over 0–8 s goes to A's unit; an output "yes" labelled A at 2–2.5 s overlaps u_1 most, belongs to A's unit, and is therefore scored there (a wrong attribution costs B's "yes" as a deletion and adds an extra word to A's unit); an output of B at 6–9 s goes to u_3; an output at 10–11 s with no speech is an insertion. Table 6 varies c.

D Hyperparameters

Simul training rows. Reads with nothing due are kept with probability 0.5; whole-window (20%) and single-speaker (15%) rows are mixed in.

Set	System	Run	0 s	0.5 s	1 s	2 s
En→Zh	Simul	windows	42.45	42.39	42.34	42.15
En→Zh	Simul	online	39.40	39.36	39.29	38.85
En→Zh	LA2	windows	27.64	27.64	27.64	27.45
En→Zh	LA2	online	33.74	33.82	33.78	33.43
Zh→En	Simul	windows	25.39	25.32	25.30	25.07
Zh→En	Simul	online	25.19	25.16	25.11	24.91
Zh→En	LA2	windows	24.73	24.73	24.69	24.46
Zh→En	LA2	online	24.41	24.38	24.34	24.11
AliMeeting	Simul	windows	12.05	12.03	12.02	11.97
AliMeeting	Simul	online	11.65	11.63	11.62	11.55
AliMeeting	LA2	windows	11.73	11.72	11.73	11.69
AliMeeting	LA2	online	10.09	10.11	10.11	10.05

Table 6: tcpBLEU on the whole-meeting subset at four collars (Simul: MIST-Simul; LA2: MIST + LA2; windows: the reference-window run of the same system, scored on the same meetings).

	MIST (offline)	MIST-Simul
Initialisation	Qwen3-Omni Thinker	MIST (offline), merged
Directions	both	one model each
Diarizer preset	offline	low latency (1.04 s)
Rows	67,795	128,879 / 160,957
Steps	2,000	3,000
GPUs (H200)	8	8
Batch (global / micro)	64 / 8	64 / 8
Learning rate	$10^{-4} \to 10^{-5}$, cosine, 50 warm-up	
LoRA	$r = 32$, $\alpha = 64$, q/k/v/o, 48 layers	
Loss weights ω	speaker 2, time 1.5, other 1	
Posterior perturbation	$p = 0.2$, $\mathcal{N}(0, 0.1^2)$, shift ≤ 1 frame	
Single-speaker rows	15%	15% (+20% offline)
Read interval / lag	–	$\Delta = 1$ s (-M: 0.5/1/2 s) / $\epsilon = 1$ s

Table 7: Training hyperparameters (MIST-Simul: En→Zh / Zh→En).

System	LAAL s	cpBLEU$_{sess}$	BLEU$_{blind}$	Spk	DER
Cascade, same-data translator	–	15.25	15.58	92.5	13.1
Cascade, turn-trained translator	–	13.05	13.11	76.4	26.0
MIST w/o speaker stream	–	8.34	16.38	87.1	52.0
+ diarizer relabelling	–	15.99	16.38	86.8	12.9
MIST (offline)	–	14.90	15.11	91.9	14.7
MIST (offline) + LA2	2.84	12.10	12.55	91.7	12.9
MIST-Simul model + LA2	3.40	13.44	13.62	91.3	10.6
MIST-Simul	3.62	11.98±1.41[2]	12.14	91.5	10.8

Table 8: Real Mandarin meetings (AliMeeting test, Zh→En, 20 meetings, 4,243 reference windows; zero-shot: no AliMeeting data in training). The English references are machine translations of the human transcripts (their automatic quality estimate: Appendix T).

Set	Seed	cpBLEU$_{sess}$	cpBLEU$_{win}$	LAAL s	Spk	Empty
En→Zh	1	42.63	43.06	2.32	90.6	0.0
	2	43.10	43.57	2.35	90.4	0.0
	3	43.52	44.01	2.53	90.3	0.0
	mean ± s.d.	43.08 ± 0.45		2.40		
Zh→En	1	24.30	25.81	2.93	93.2	0.0
	2	22.79	24.07	3.16	93.0	0.0
	mean ± s.d.	23.55 ± 1.07		3.04		
AliMeeting	1	12.97	13.09	3.09	91.5	0.0
	2	10.98	11.08	4.15	91.5	0.0
	mean ± s.d.	11.98 ± 1.41		3.62		

Table 9: MIST-Simul per training seed (Δ=1 s, reference windows; AliMeeting uses the Zh→En models). Seed 1 is the model of v6; seeds 2 and 3 differ only in the random seed (seed 3 of En→Zh was retrained from scratch, prereg_v7 amendment 6). Empty: windows with no output (failures).

E Latency and Full Metrics

Read interval. Varying Δ at test time (Figure 3): at 2 s, MIST-Simul loses 1.97 points for 0.29 s more LAAL (LocalAgreement trails by 3.37); at 0.5 s, earlier than any training read, it commits guesses and collapses (23.31; Zh→En 5.89). Drawing the interval per training window from $\{0.5, 1, 2\}$ s (MIST-Simul-M) removes the collapse: 42.79 at 0.5 s with 2.18 s LAAL, by waiting longer, not by sub-second commits.

F Statistics

Every comparison uses one procedure. The difference A−B is taken over the windows both systems decoded. Its 95% interval comes from 1,000 bootstrap resamples of whole meetings (16 AMI, 20 AliMeeting); the slot-to-speaker matching is made per meeting, so a meeting resample re-uses exactly the matching a re-match would find. The p value is an exact two-sided permutation test that flips the two systems' outputs per meeting over all 2^{16} (AMI) or 2^{20} (AliMeeting) assignments, so it cannot go below $2^{-15} \approx 3 \cdot 10^{-5}$ or 2^{-19}. Leave-one-meeting-out minima and maxima and the per-meeting differences are in the released statistics file. Seed comparisons average the difference over the training seeds; their bootstrap resamples meetings and seeds, their permutation flips a meeting's outputs for all seeds at once. The decoder adds noise that neither test contains: greedy bf16 decoding of the MoE Thinker is not batch-invariant (Appendix L), so we widen the meeting interval by $\sigma_{\text{dec}}(n) = \sigma_0/\sqrt{n}$ for n scored windows (variances added), with σ_0 the larger of the v6 estimate (12.3) and the v7 re-decode (−0.09 tcpBLEU on 1,412 windows). A difference is called real only if $p < 0.05$ (Holm-adjusted over the eight preregistered tests of Table 13) and the widened interval excludes 0. Table 14 lists the exploratory comparisons without correction.

G Single- and Multi-Speaker Windows, Errors

Remaining errors. Most remaining errors are short back-channels ("yeah", "okay") of a second speaker in overlap that are merged into the main turn or dropped: 25.9% of the multi-speaker

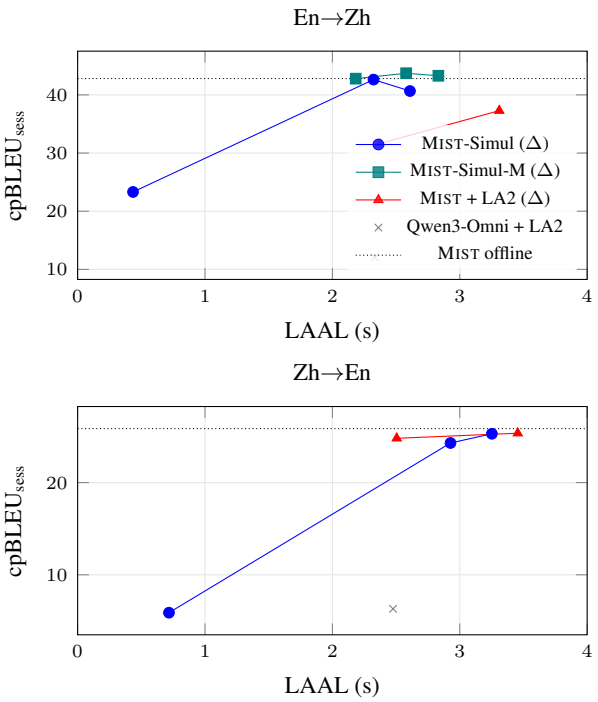


Figure 3: Latency–quality trade-off (text) over the read interval $\Delta \in \{0.5, 1, 2\}$ s (left to right). MIST-Simul is trained at $\Delta = 1$ s, MIST-Simul-M at mixed intervals (En→Zh). The dotted line is the offline MIST. Numbers in Table 10.

En→Zh windows are under-counted (41.2% without the speaker stream), while whole utterances are rarely omitted (2.6%; Appendix G).

MIST loses 47.42 → 39.39 cpBLEU$_{sess}$ from single- to multi-speaker windows (En→Zh) and finds the right number of speakers in only 72.8% of the latter. Table 15 splits the test set by the number of speakers, Table 17 counts the errors on multi-speaker windows after aligning each output segment to a reference utterance by time overlap and text similarity, and Table 16 evaluates on windows cut by the diarizer.

H Example

Table 18 shows a window in which speaker B says "yes" while A is still talking. The cascade translates the overlapped stretch in every turn that contains it: "yes" appears in A's turn, in a spurious third slot and in B's turn. MIST writes it once and attributes it to B.

I Chinese AMI Meetings

Construction. The Chinese meetings reuse the En→Zh translations of every AMI utterance. A first version laid the MT pipeline's own Chinese speech (one voice clone per utterance) on the meeting clock; the audit below found that the diarizer split most speakers into several slots, because a person sounded different from one utterance to the next. We therefore re-spoke every utterance with the UniSS Talker and one voice prompt per speaker. Each clip is trimmed of leading and trailing silence, set to its speaker's median loudness in the English recording (gain at most $4\times$, peak below 0.9), placed at the utterance's original start, and sped up or slowed down (pitch kept) towards the English duration within $[0.8, 1.5]$; a speaker's next clip waits until the previous one ends. The meeting's background is the quieter half of its speech-free English frames.

Audit. Table 19 lists the checks, the problems they found and the fixes; every check was rerun after its fix. An utterance that fails the clip checks is left out of the meeting (neither heard nor translated) rather than dropping its window, which would remove most multi-speaker windows, since the failures are mostly short back-channels.

J Read Targets

Deadline targets (§3.7) are derived from the reference and its timing. An alternative that needs neither is to label each read with the *stable prefix* of the model's own output: the offline model drafts

System	En→Zh					Zh→En				
	LAAL	First	cpBLEU$_{sess}$	BLEU$_{blind}$	Len	LAAL	First	cpBLEU$_{sess}$	BLEU$_{blind}$	Len
Qwen3-Omni + LA2, Δ=1 s	2.33	1.85	12.07	26.87	0.75	2.48	2.08	6.31	14.11	0.78
MIST (offline) + LA2, Δ=1 s	2.37	2.23	31.64	32.74	1.18	2.50	2.24	24.81	27.67	0.94
MIST (offline) + LA2, Δ=2 s	3.31	2.98	37.29	38.33	1.06	3.46	2.98	25.36	28.21	0.94
MIST-Simul, Δ=0.5 s	0.43	0.25	23.31	24.30	1.27	0.72	0.24	5.89	10.79	1.34
MIST-Simul, Δ=1 s	2.32	2.19	42.63	42.80	0.99	2.93	2.79	24.30	26.71	1.03
MIST-Simul, Δ=2 s	2.61	2.43	40.66	40.86	1.01	3.25	3.03	25.30	27.71	0.94
MIST-Simul-M, Δ=0.5 s	2.18	2.23	42.79	42.97	0.98			–		
MIST-Simul-M, Δ=1 s	2.58	2.45	43.72	43.78	0.97			–		
MIST-Simul-M, Δ=2 s	2.83	2.67	43.29	43.47	0.97			–		
MIST-Simul, 1,000 steps			–			2.85	2.83	24.78	27.26	0.96

Table 10: Latency and quality of the simultaneous systems at read intervals Δ (MIST-Simul is trained with reads every 1 s; MIST-Simul-M with reads every 0.5, 1 or 2 s, drawn per training window, En→Zh only; the last row is the 1,000-step Zh→En model of our previous version; Zh→En at 0.5 s: 1,419 of the 3,317 windows were decoded with batch 4 instead of 16 after a memory stop). LAAL and *First* (delay of the first commit after the first source utterance starts) in seconds. Len: output over reference length (characters for Chinese, words for English).

System	cpBLEU$_{sess}$	cpBLEU$_{win}$	cpchrF$_{sess}$	BLEU$_{blind}$	Len	Spk	DER	Conf.	Empty
Qwen3-Omni, prompted	12.82	26.91	12.50	29.01	0.77	65.0	67.6	46.8	0.0
Cascade: diarizer → Qwen3-Omni	29.47	29.52	25.07	29.16	0.87	88.2	25.8	0.8	0.0
Cascade: diarizer → same-data model	36.29	36.36	33.90	37.01	1.11	88.2	25.8	0.8	0.0
MIST w/o speaker stream	21.17	42.32	19.65	43.40	0.95	84.8	54.5	43.5	0.1
MIST (offline)	42.81	42.97	36.71	43.33	0.95	89.5	14.0	1.7	0.4
MIST (offline), oracle stream	42.32	42.58	36.27	42.95	0.95	92.2	14.2	1.3	0.3
Qwen3-Omni + LA2	12.07	25.05	11.86	26.87	0.75	65.0	101.5	0.1	0.0
MIST (offline) + LA2	31.64	32.16	33.52	32.74	1.18	87.8	15.2	3.9	0.0
MIST-Simul	42.63	43.06	36.81	42.80	0.99	90.6	11.0	2.3	0.0

Table 11: All text metrics, En→Zh. cpBLEU$_{win}$ matches speakers once per window; Len: output over reference length (characters); Conf.: speaker-confusion part of DER; Empty: windows with no output.

a continuation from the audio heard so far and another from the whole window, both after the same committed prefix, and the target is the longest common prefix of the two (speaker and time tokens included), followed by the wait token. We continued the same offline model for 3,000 steps on such reads of 8,000 training windows (both directions, intervals drawn from $\{0.5, 1, 1, 2\}$ s) and decode it with the same plain wait rule (greedy, the wait token competing with all others). Table 20 compares the two at $\Delta = 1$ s: deadline targets are far better in En→Zh (9.90 cpBLEU$_{sess}$, $p < 0.001$) and slightly better but later in Zh→En (0.23, not significant), so we use them.

K Causality of the Speaker Stream

The streaming runs read posteriors from one low-latency run of the diarizer over the whole session, with every frame later than $t - \lambda$ zeroed at a read at t (Eq. 12). This equals a causal system only if those frames do not change when the audio after t is never given. For every test meeting of the three sets we drew three reference windows and one read time in each (fixed seed), truncated the session audio at that time and ran the diarizer from scratch with the same model and preset, pooling its 10 ms output to 80 ms exactly as for the stream. Over 156 cuts and 1,953,350 compared frames, the largest absolute difference to the posteriors the streaming runs read was 4.9e-04 (the fp16 precision in which they are stored), the mean absolute difference 3.7e-06, the most active slot agreed on at least 99.94% and the whole set of active slots on at least 99.96% of active frames in every cut, and the > 0.5 decision differed on 20 frames (9 of them among the 197,598 overlapped frames), every one of them a posterior within $5 \cdot 10^{-4}$ of 0.5, i.e. an fp16 rounding tie. Our pre-registered criterion (all decisions identical) is therefore missed by these rounding ties; we consider the stream causal up to floating-point noise and did not re-decode with prefix posteriors. Prefix runs of the same meeting cut at different times also agree with each other (104 pairs, largest difference 2.4e-04): later audio never changes earlier frames. (A first comparison script compared the 10 ms prefix output with the 80 ms stream and reported large differences; it was a frame-rate error, corrected before

System	cpBLEU$_{\text{sess}}$	cpBLEU$_{\text{win}}$	cpchrF$_{\text{sess}}$	BLEU$_{\text{blind}}$	Len	Spk	DER	Conf.	Empty
Qwen3-Omni, prompted	6.40	13.55	19.00	14.24	0.76	63.5	78.7	50.6	0.0
Cascade: diarizer → Qwen3-Omni	16.07	16.67	40.61	16.84	0.94	83.6	15.5	4.1	0.0
Cascade: diarizer → same-data model	22.36	23.22	48.18	24.85	1.04	83.6	15.5	4.1	0.0
MIST w/o speaker stream	12.20	26.24	26.17	28.13	0.91	84.9	57.4	48.5	0.1
MIST (offline)	25.86	27.06	49.06	28.43	0.91	91.3	16.0	5.4	0.0
MIST (offline), oracle stream	27.32	27.39	51.29	28.37	0.92	95.9	11.6	1.2	0.1
Qwen3-Omni + LA2	6.31	13.41	18.74	14.11	0.78	63.5	101.6	0.1	0.0
MIST (offline) + LA2	24.81	26.23	47.98	27.67	0.94	90.7	17.3	6.6	0.0
MIST-Simul	24.30	25.81	47.72	26.71	1.03	93.2	12.8	6.3	0.0

Table 12: All text metrics, Zh→En. cpBLEU$_{\text{win}}$ matches speakers once per window; Len: output over reference length (words); Conf.: speaker-confusion part of DER; Empty: windows with no output.

Test	Comparison	Mtgs	Δ	95% CI	+ decoder noise	p_{perm}	p_{Holm}	verdict
E-on1	En→Zh whole meetings, MIST-Simul vs. MIST + LA2 (tcpBLEU)	16	5.83	[3.09, 8.48]	[3.17, 8.48]	<0.001	<0.001	a better
E-on2	Zh→En whole meetings, same	16	0.59	[−1.00, 1.15]	[−0.56, 1.75]	0.079	0.238	tie
E-on3	AliMeeting whole meetings, same	20	0.38	[−1.55, 2.19]	[−1.52, 2.28]	0.711	0.951	tie
E-cas1	En→Zh, MIST vs. turn-trained cascade (cpBLEU$_{\text{sess}}$)	16	7.54	[4.86, 10.47]	[4.70, 10.38]	<0.001	<0.001	a better
E-cas2	Zh→En, same	16	2.17	[1.65, 2.81]	[1.44, 2.90]	<0.001	<0.001	a better
E-cas3	AliMeeting, same	20	1.85	[1.14, 2.63]	[1.03, 2.67]	<0.001	0.001	a better
E-seed-en	En→Zh, MIST-Simul (seed mean) vs. MIST + LA2	16	11.44	[4.45, 17.22]	[5.09, 17.79]	<0.001	0.003	a better
E-seed-zh	Zh→En, same	16	−1.27	[−4.75, 0.73]	[−4.22, 1.69]	0.475	0.951	tie

Table 13: The eight pre-registered tests of v7 (prereg_v7). Δ: first minus second system; 95% CI: cluster bootstrap over meetings; + decoder noise: the same interval widened by the run-to-run noise of decoding (Appendix F); p_{perm}: exact two-sided permutation test over meeting sign flips; p_{Holm}: Holm-adjusted over the eight. A difference is called real only if $p_{\text{Holm}} < 0.05$ and the widened interval excludes 0; otherwise "tie".

the numbers above.) The audio encoder needs no such test: each read encodes the audio prefix itself. Algorithm 4 lists one read.

Algorithm 1: read r of a window that started at s (meeting clock)

1: $t \leftarrow \min(r\Delta, T)$; audio $\boldsymbol{x}_{s:s+t}$ (only what has been heard)
2: $\boldsymbol{A} \leftarrow \text{Enc}(\boldsymbol{x}_{s:s+t})$ (prefix forward, Eq. 2)
3: $\boldsymbol{P}^{(r)} \leftarrow$ diarizer frames of $(s, s+t-\lambda]$, later ones zero (Eq. 12)
4: $\boldsymbol{E}^{(r)} \leftarrow$ interleave $\boldsymbol{A}$ with speaker tokens of $\boldsymbol{P}^{(r)}$ (Eq. 7)
5: greedy decode after the committed tokens until ⟨wait⟩ (or ⟨eos⟩ if $t = T$)
6: commit the new tokens; parse segments; hand each new block to the Talker
7: whole meetings: commit time ← max(now, window opening known); at $t = T$, max(·, window end known) (§3.9)
no slot active ⇒ no window; empty decode ⇒ nothing spoken

Figure 4: One read of MIST-Simul. The full prompt is re-encoded at every read (no cache is kept across reads).

L Decoder Run-to-Run Noise

Batch invariance. Greedy decoding of the MoE Thinker in bf16 is not invariant to the batch composition: decoding the same 96 Zh→En windows at batch 16 and at batch 4 gives token-identical output for only 38.5% of them, with cpBLEU$_{\text{sess}}$ 25.10 vs. 25.56. Differences of this size between runs decoded in different batches (including the shards re-decoded after a stopped worker, and the late runs that split the Thinker over two cards, prereg_v6 amendment 6) are within this noise. To size it for our tests, we resample N windows with replacement from the 96 and take the spread of the difference between the two runs: its standard deviation is 0.32 cpBLEU at N=1,410 (a whole-meeting subset), 0.25 at 2,375 and 0.20 at 3,317. Neither the meeting bootstrap nor the permutation test contains this source: they treat each system's output as fixed. We therefore widen every interval by it (Appendix F) and call a difference real only if the widened interval excludes 0. A second estimate comes from re-decoding the 8 En→Zh meetings decoded in v6 with the v7 decoder (Appendix Q): −0.09 tcpBLEU on 1,412 windows, smaller than the first; we keep the larger.

M Deadline Targets: How Accurate?

For 2,000 validation utterances per direction we aligned source and translation with a SimAlign-style aligner (Jalili Sabet et al., 2020) (multilingual BERT, layer 8, mutual argmax) and took, for every chunk, the end of the last source word or character aligned to it or to an earlier chunk, i.e. what must have been heard to write it. The En→Zh rule (character fraction mapped to AMI word times) deviates from this by 0.17 s on average; 6.5% of its deadlines precede the evidence, and 1.7% do so by more than the 1 s lag. The Zh→En duration rule of v5 deviates by 0.24 s (15.0% early, 0.4% by more than the lag); with forced-aligned character times (FunASR timestamp predictor (Gao et al., 2023)) it deviates by 0.19 s (3.5% early). Because a win-

Set	Comparison	Mtgs	Δ	95% CI	+ decoder noise	p_{perm}	verdict
En→Zh	MIST vs. same-data casc.	16	6.52	[5.47, 7.59]	[5.36, 7.68]	<0.001	a better
En→Zh	MIST vs. relabelling	16	0.75	[0.13, 1.31]	[−0.02, 1.52]	0.036	tie
En→Zh	MIST-Simul vs. MIST+LA2	16	10.98	[3.87, 16.73]	[4.53, 17.44]	0.006	a better
En→Zh (speech)	MIST vs. same-data casc.	16	1.63	[0.26, 2.90]	[0.23, 3.04]	0.048	a better
En→Zh (speech)	MIST-Simul vs. MIST+LA2	16	2.10	[1.47, 2.69]	[1.31, 2.89]	<0.001	a better
Zh→En	MIST vs. same-data casc.	16	3.50	[2.68, 4.36]	[2.54, 4.46]	<0.001	a better
Zh→En	MIST vs. relabelling	16	1.20	[0.81, 1.56]	[0.64, 1.75]	<0.001	a better
Zh→En	MIST-Simul vs. MIST+LA2	16	−0.52	[−2.70, 0.66]	[−2.45, 1.41]	0.535	tie
Zh→En (speech)	MIST vs. same-data casc.	16	1.97	[0.59, 3.49]	[0.49, 3.45]	0.014	a better
Zh→En (speech)	MIST-Simul vs. MIST+LA2	16	2.00	[−0.30, 4.78]	[−0.52, 4.52]	0.154	tie
AliMeeting	MIST vs. same-data casc.	20	−0.35	[−1.78, 1.08]	[−1.85, 1.14]	0.458	tie
AliMeeting	MIST vs. relabelling	20	−1.09	[−1.54, −0.68]	[−1.67, −0.52]	<0.001	b better
AliMeeting	MIST-Simul vs. MIST+LA2	20	0.87	[−0.54, 2.07]	[−0.51, 2.25]	0.276	tie
AliMeeting (speech)	MIST vs. same-data casc.	20	−0.75	[−2.12, 0.49]	[−2.15, 0.65]	0.115	tie
AliMeeting (speech)	MIST-Simul vs. MIST+LA2	20	1.21	[−0.22, 2.20]	[−0.08, 2.50]	0.106	tie
En→Zh	MIST vs. turn-trained casc.	16	7.54	[4.86, 10.47]	[4.70, 10.38]	<0.001	a better
En→Zh	turn-trained casc. vs. same-data casc.	16	−1.02	[−3.72, 1.64]	[−3.74, 1.70]	0.304	tie
En→Zh	MIST vs. ref.-turn casc.	16	5.38	[4.27, 6.42]	[4.17, 6.60]	<0.001	a better
En→Zh	MIST vs. context casc.	16	19.06	[18.11, 19.78]	[18.10, 20.01]	<0.001	a better
Zh→En	MIST vs. turn-trained casc.	16	2.17	[1.65, 2.81]	[1.44, 2.90]	<0.001	a better
Zh→En	turn-trained casc. vs. same-data casc.	16	1.33	[0.15, 2.46]	[0.08, 2.57]	0.003	a better
Zh→En	MIST vs. ref.-turn casc.	16	0.91	[−0.22, 1.98]	[−0.29, 2.12]	0.208	tie
Zh→En	MIST vs. context casc.	16	8.98	[7.93, 10.01]	[7.88, 10.08]	<0.001	a better
AliMeeting	MIST vs. turn-trained casc.	20	1.85	[1.14, 2.63]	[1.03, 2.67]	<0.001	a better
AliMeeting	turn-trained casc. vs. same-data casc.	20	−2.20	[−3.81, −0.50]	[−3.97, −0.44]	0.004	b better
En→Zh	MIST-Simul vs. stream. casc.	16	7.47	[5.57, 9.37]	[5.50, 9.44]	<0.001	a better
En→Zh	MIST-Simul vs. Simul model+LA2	16	3.42	[0.14, 8.08]	[−0.88, 7.72]	0.134	tie
En→Zh	Simul model+LA2 vs. MIST+LA2	16	7.57	[1.72, 13.57]	[1.55, 13.58]	0.039	a better
En→Zh	MIST-Simul vs. fixed lag 1 s	8	23.29	[20.78, 25.38]	[20.84, 25.73]	0.008	a better
En→Zh	MIST-Simul vs. fixed lag 2 s	8	−0.75	[−2.91, 0.63]	[−2.78, 1.28]	0.148	tie
En→Zh	fixed lag 2 s vs. MIST+LA2	8	15.35	[3.36, 21.89]	[5.85, 24.86]	0.008	a better
Zh→En	MIST-Simul vs. stream. casc.	16	2.74	[0.54, 4.19]	[0.88, 4.61]	0.010	a better
Zh→En	MIST-Simul vs. Simul model+LA2	16	−0.56	[−2.74, 0.55]	[−2.41, 1.29]	0.594	tie
Zh→En	Simul model+LA2 vs. MIST+LA2	16	0.04	[−0.32, 0.45]	[−0.53, 0.62]	0.825	tie
AliMeeting	MIST-Simul vs. Simul model+LA2	20	−0.47	[−1.74, 0.49]	[−1.75, 0.81]	0.548	tie
AliMeeting	Simul model+LA2 vs. MIST+LA2	20	1.34	[−0.31, 2.44]	[−0.08, 2.76]	0.089	tie
En→Zh whole mtgs	sp2_on_S2dl vs. sp2_on_O2_LA2	16	1.58	[1.15, 1.97]	[1.00, 2.16]	<0.001	a better
Zh→En whole mtgs	sp2_on_S2dl vs. sp2_on_O2_LA2	16	1.69	[0.02, 3.47]	[−0.15, 3.53]	0.096	tie
AliMeeting whole mtgs	sp2_on_S2dl vs. sp2_on_O2_LA2	20	0.29	[−1.03, 1.48]	[−1.02, 1.61]	0.671	tie

Table 14: Exploratory comparisons of v7 ($\text{cpBLEU}_{\text{sess}}$ on reference windows, ASR-cpBLEU for speech, tcpBLEU for whole meetings), on the windows both systems decoded. Columns as Table 13; no multiple-test correction (exploratory). Fixed lag is on the 8-meeting subset (prereg_v7 amendment 2).

System	$\text{cpBLEU}_{\text{sess}}$ single	$\text{cpBLEU}_{\text{sess}}$ multi	DER single	DER multi	Spk-count multi
En→Zh					
Qwen3-Omni, prompted	16.64	9.97	68.0	65.4	0.0
Cascade: diarizer → same-data model	43.21	31.97	22.1	28.6	78.7
MIST w/o speaker stream	25.04	18.28	48.1	57.7	57.6
MIST (offline)	47.42	39.39	6.9	19.6	72.8
MIST (offline), oracle stream	47.59	38.40	6.2	20.5	80.1
MIST (offline) + LA2	43.75	25.42	8.4	20.5	68.1
MIST-Simul	45.79	40.21	5.4	15.4	77.0
Zh→En					
Qwen3-Omni, prompted	9.26	4.49	79.6	75.1	0.0
Cascade: diarizer → same-data model	27.71	19.34	13.5	16.9	74.2
MIST w/o speaker stream	15.68	9.94	50.0	60.9	59.6
MIST (offline)	29.09	23.76	9.9	20.3	77.7
MIST (offline), oracle stream	30.64	25.16	4.9	16.4	88.8
MIST (offline) + LA2	27.07	23.36	13.2	20.2	75.6
MIST-Simul	27.61	21.64	10.2	14.6	82.3

Table 15: Single- and multi-speaker test windows.

dow ends 0.3 s after its last word and the lag is 1 s, 34.7% of Zh→En training chunks are due only at the last read; at test time MIST-Simul commits 34.0% (En→Zh) and 39.8% (Zh→En) of its output at the last read of a window (LA2 on MIST:

System	$\text{cpBLEU}_{\text{sess}}$	$\text{BLEU}_{\text{blind}}$	Spk-count	DER
Qwen3-Omni, prompted	14.85	29.34	61.2	63.4
Cascade: diarizer → same-data model	37.53	37.91	86.3	20.7
MIST w/o speaker stream	21.57	43.91	83.8	55.4
MIST (offline)	43.36	43.83	89.4	15.5

Table 16: En→Zh test windows opened and closed by the diarizer itself (871 windows that contain whole utterances), offline.

37.4%), which is why the whole-meeting evaluation moves those commits to the moment the end is known.

N Segment-Level Attribution

Table 21 scores who-said-what-when per reference utterance. The cascades' turns come straight from the diarizer, so their time placement is as good as or better than MIST's; MIST's gain in cp-

System	Under	Over	Wrong	Omit	Insert	Start s
En→Zh						
MIST w/o speaker stream	41.2	1.2	77.4	2.4	0.9	0.42
MIST (offline)	25.9	1.3	10.9	2.6	4.2	0.42
MIST (offline), oracle stream	19.1	0.7	11.7	4.4	3.9	0.42
MIST (offline) + LA2	29.6	2.3	13.5	1.6	3.8	0.48
MIST-Simul	21.1	1.9	14.3	0.3	5.9	0.82
Zh→En						
MIST w/o speaker stream	38.9	1.5	71.0	1.2	1.5	0.26
MIST (offline)	18.2	4.1	17.2	1.6	3.3	0.28
MIST (offline), oracle stream	11.2	0.0	11.1	1.2	2.9	0.28
MIST (offline) + LA2	19.8	4.6	17.9	1.1	3.8	0.29
MIST-Simul	14.5	3.1	17.4	0.2	6.1	0.42

Table 17: Errors on the multi-speaker test windows (%). *Under*/*Over*: windows with too few / too many speakers; *Wrong*: segments whose speaker differs from that of the reference utterance they align to; *Omit*: reference utterances no segment aligns to; *Insert*: segments aligned to no reference; *Start*: mean start-time error (s), per committed chunk for the simultaneous systems. Slots are those the targets were built with (offline or low-latency preset). Without the speaker stream, *Wrong* is measured against slots the model never sees.

BLEU is a translation gain on the same attribution, not better segmentation. Recall on a speaker's first utterance after $\geq$60 s of silence (re-entry) stays within a few points of the overall recall for every system that reads the diarizer, and drops by half without it: identity over the meeting is the diarizer's contribution.

O Voice Prompts

A causal voice prompt ($\geq$1 s of the slot heard alone before its first commit) exists for 97.6% (En→Zh), 97.5% (Zh→En) and 98.9% (AliMeeting) of the output speaker streams of MIST-Simul, with 7.6 s of audio on average; during the prompt the strongest other slot has a mean posterior of 0.027 (En→Zh) and 0.072 (Zh→En). The rest (2.4% / 2.5% / 1.1%) fall back to the window's audio up to the first commit, which may contain other voices. Cold start (re-review M17): from a slot's first activity it takes a median 9.0 s (En→Zh), 8.6 s (Zh→En) and 1.6 s (AliMeeting) until 1 s of it has been heard alone; segments committed earlier use the fallback. Every test meeting has fewer than eight speakers, so the slots never run out; larger meetings are untested. Voice confusion (Table 22) counts only output streams of windows with at least two reference speakers; single-speaker windows have no other voice to confuse.

P Pre-registration and Amendments

All v7 analyses were registered in `prereg_v7` before any v7 decode or score; every change was committed as a dated amendment before the results it touches: registration `165dbf4` (2026-11-01 14:21 UTC); amendment 1 `4ebc26c` (2026-11-01 14:32 UTC); amendment 2 `1a0f35b` (2026-11-01 14:35 UTC); amendment 3 `3f6b23b` (2026-11-02 03:38 UTC); amendment 4 `305c19a` (2026-11-02 06:43 UTC); amendment 5 `15ce328` (2026-11-02 09:03 UTC); amendment 6 `9982c5a` (2026-11-02 13:29 UTC); amendment 7 `c6d5a7a` (2026-11-02 13:44 UTC); amendment 8 `2c3ef0c` (2026-11-02 13:45 UTC); amendment 9 `e37534b` (2026-11-03 00:13 UTC). The registration fixes the full meeting lists (AMI test EN2002a–d, ES2004a–d, IS1009a–d, TS3003a–d; all 20 AliMeeting test meetings), the eight confirmatory tests and their Holm family, the decoder-noise widening and the dev gates. The amendments were forced by compute and by colleagues' jobs on the shared cards: they cut the K, loss-weight and noise ablations and the matched-data deadline/stable-prefix control (amendment 2), restricted fixed lag to the 8-meeting En→Zh subset (amendment 2), retrained the third En→Zh seed from scratch after a resume restarted its learning-rate schedule (amendments 3, 6), and changed decoding layouts and batch sizes (amendments 4, 5, 7, 8), which only enter through decoder noise. The v6 subset results that v7 extends were registered in `prereg_v6` and are reused unchanged.

Q Whole-Meeting Runs and Listener-Side Lag

Online procedure. For each meeting, with $\lambda = 1.04$ s the diarizer's look-ahead:

1. The streaming diarizer runs over the whole meeting once; its frame f is final at $t_f + \lambda$ and never revised (Appendix K). Slots persist for the whole meeting.
2. A window opens at the first final frame with any slot > 0.5 and closes after 1 s without an active slot or when it reaches 20 s; the opening is known at its frame $+\lambda$, the end at the deciding frame $+\lambda$. The window's audio is $[\text{open} - 0.2\,\text{s},\ \text{last active frame} + 0.3\,\text{s}]$; the 0.2 s before the opening are kept in a ring buffer of the past audio. A window cut at 20 s is followed by a new window that opens

English source	**A** [0.3–4.3 s] uh so this is these are the circuits un underneath the buttons **A** [4.3–6.9 s] uh can you see the black uh round marks **A** [6.9–8.1 s] they they are the circuits **B** [6.9–7.2 s] yes
Reference	**A** [0.3] 呃，这些就是按钮下面的电路。 **A** [4.3] 呃，你能看到那些黑色的圆形标记吗？ **A** [6.9] 它们就是电路 **B** [6.9] 是的
Cascade (same data)	**A** [0.2] 呃，这些就是按钮下面的电路。 **A** [4.3] 呃，你能看到那些黑色的圆形标记吗？它们就是电路。是的？ [6.6] 是的 **B** [6.9] 是的
MIST (offline)	**A** [0.3] 呃，这些就是按钮下方的电路。 **A** [4.4] 呃，你能看到那些黑色的圆形标记吗？它们就是电路。 **B** [7.0] 是的

Table 18: Test window `IS1009c_0126` with overlapping speech. Speakers are lettered by first appearance in the reference; output speakers are mapped to reference speakers by the meeting-level matching of $\text{cpBLEU}_{\text{sess}}$ ("?" = unmatched). Times in seconds from the window start.

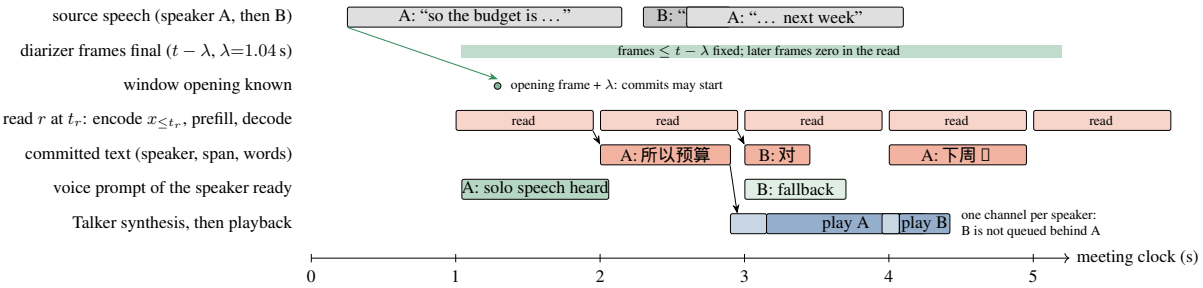


Figure 5: One online window, 0–5 s, to scale except for the text. Audio arrives continuously; a read at t_r (every 1 s) re-encodes the audio prefix $x_{\leq t_r}$ from scratch, reads diarizer frames up to $t_r - \lambda$ (later frames are zero), and decodes until $\langle\text{wait}\rangle$; with batch 1 on the two-card Thinker a read takes about 1 s, so its text is committed at about $t_r + 1$ s (the measured median, Appendix Q). Nothing is committed before the window's opening is known (λ after its first active frame). The Talker speaks a block once it is committed and the speaker has a voice prompt (its solo frames heard before the first commit, Eq. 17; else a fallback); synthesis of the next block runs while the previous one plays. With one channel for all speakers, B's "yes" would wait until A's block ends.

Check	Problem found	Fix
Overlap vs. English (±20%)	+36% (Chinese longer)	speed-up $\leq 1.5\times$
	-25% after re-speaking	slow-down $\leq 1.25\times$: -12 to -17%
Windows	clock shifts split groups	English utterance groups
Intelligibility (CER ≤ 0.3)	17% fail Whisper	Paraformer recheck; failing clips left out
Clipping	bursts in background	quieter frames only, gain cap
Diarizer (DER $\leq$ English + 10)	40–44% vs. 28%, extra slots	one voice per speaker: 17.7–21.7%
	13 meetings still fail	meetings dropped (none in test)

Table 19: Audit of the Chinese AMI meetings. Left out in the end: 19,081 of 123,353 utterances (CER 13,681; no translation 5,295; long pauses 98; other 7).

	LAAL	$\text{cpBLEU}_{\text{sess}}$	Units/blk	1-unit %	Mid %
En→Zh					
Deadline labels (MIST-Simul)	2.32	42.63	6.0	11.1	12.9
Stable-prefix labels	2.43	32.73	8.3	8.2	36.9
Zh→En					
Deadline labels (MIST-Simul)	2.93	24.30	5.3	36.9	0.6
Stable-prefix labels	2.35	24.07	4.2	32.8	32.3

Table 20: Read targets at $\Delta = 1$ s, both continued from the same offline model for 3,000 steps. Stable-prefix labels: at each read, the prefix of the model's own draft that it also writes after hearing the whole window, decoded with the same plain wait rule. Units/blk: mean characters (Zh) or words (En) per committed block; 1-unit: blocks of a single unit; Mid: blocks that stop mid-phrase before the same speaker's next block.

at the next active frame, so a sentence spoken across the cut is split and translated in two parts (Table 24: coverage of 20 s-cut windows).

3. Reads at $t_r = r\Delta$ within the window: the encoder re-encodes $x_{\leq t_r}$ from scratch, the speaker stream holds the final frames $\leq t_r - \lambda$ (later frames zero, Eq. (12)), and the decoder, restarted at every window, continues greedily from its own committed tokens until $\langle\text{wait}\rangle$. No key-value cache is kept across reads.

4. Commit times are clamped to the window's detected opening; the last read, which writes until $\langle\text{eos}\rangle$, commits at the moment the window's end is known. Committed text is never revised; the 0.2 s / 0.3 s padding can make two windows translate the same short utter-

Set	System	P	R	F1	Omit	Misattr.	Onset	Offset	Re-entry	≤1 s
En→Zh	Prompted	50.4	36.6	42.4	0.0	63.4	1.33	1.28	20.4	30.6
	Cascade, same data	94.8	94.3	94.6	0.4	5.3	0.04	0.17	89.1	86.6
	w/o speaker stream	56.8	51.0	53.8	1.0	48.0	0.05	0.16	40.1	43.8
	MIST	97.5	91.1	94.2	1.2	7.8	0.05	0.18	83.3	82.1
	MIST-Simul	94.5	87.5	90.9	0.0	12.4	0.03	0.07	76.5	68.7
	MIST + LA2	91.0	85.3	88.0	0.1	14.6	0.06	0.23	75.5	65.8
	MIST-Simul-M	94.7	87.4	90.9	0.0	12.6	0.03	0.07	75.5	68.7
Zh→En	Cascade, same data	87.2	91.6	89.3	0.5	7.8	0.06	0.06	84.1	85.1
	w/o speaker stream	54.8	49.6	52.1	0.5	49.9	0.04	0.05	41.2	45.3
	MIST	93.5	89.6	91.5	0.4	10.0	0.03	0.06	81.6	81.9
	MIST-Simul	90.6	86.9	88.7	0.0	13.1	0.02	0.03	75.5	74.3
	MIST + LA2	89.2	85.3	87.2	0.1	14.6	0.04	0.08	70.8	70.4
AliMeeting	Cascade, same data	94.1	97.9	96.0	0.1	2.0	0.10	0.12	91.4	91.8
	w/o speaker stream	64.3	60.1	62.1	1.1	38.8	0.14	0.21	53.0	55.9
	MIST	96.5	93.0	94.7	2.8	4.2	0.09	0.18	83.5	83.1
	MIST-Simul	95.2	92.8	94.0	0.1	7.1	0.07	0.11	81.7	72.3
	MIST + LA2	93.9	91.6	92.7	0.1	8.3	0.09	0.20	81.7	70.5

Table 21: Segment-level attribution on the test windows. Each reference utterance is checked against the output segments overlapping it (0.5 s collar): recall (R) = covered by its own speaker after the session matching, Omit = no segment at all, Misattr. = only other speakers' segments; precision (P) = segments that overlap an utterance of their matched speaker. Onset / offset: median error of the nearest segment (s). Re-entry: recall on utterances of a speaker silent for ≥60 s before; ≤1 s: recall on utterances of at most one second.

ance ("repeat" in Table 24); we do not deduplicate.

5. Each committed block is spoken by the Talker with the speaker's causal prompt (Eq. (17)) and placed on the channel by Eq. (18): one channel for all speakers, or one per speaker slot.

The online windows cover all speech: 3,387 (En→Zh), 3,365 (Zh→En) and 2,837 (AliMeeting) windows. The 8+8+10 meetings decoded in v6 are reused unchanged; the others were decoded in v7 with the Thinker split over two cards or on one card (prereg_v7 amendments 4–8). Re-decoding the 8 En→Zh meetings of v6 with the v7 decoder gave 38.76 vs. 38.85 tcpBLEU (34.8% of windows token-identical), the decoder noise of Appendix F.

Listener-side lag. Every committed block gets its commit time plus the median single-stream time of one read with the Thinker on two H200 cards (batch 1, HuggingFace eager decoding; 1008 ms for MIST-Simul, 2028 ms for LA2, which decodes to the end at every read), plus synthesis at the Talker's measured single-stream rate (161.80 codec tokens/s), and is played at the later of that time and the end of the previous block on its channel (0.2 s more on a change of speaker). Synthesis of a block can overlap the next read; reads of one window are sequential. Output streams of more than 400 blocks (runaway repetition) were not synthesised (prereg_v7 amendment 1): they count as empty speech, and both systems' listener lag is reported without them. One channel serialises overlapping speech and so builds a backlog (Table 29); one channel per speaker keeps it near zero.

R Coverage of the Reference Windows

Table 25 follows the test speech from the meetings to the reference windows of Tables 2 and 8: windows are reference utterance groups that touch or overlap, kept only if they last at most 20 s, do not straddle another group and pass the automatic translation and synthesis review (Appendix A). Overlapped speech is kept least. The whole-meeting runs of §5.4 are scored on the same reference utterances but read all speech; En→Zh loses 2.36 tcpBLEU against the reference windows of the same meetings, Zh→En none.

Dir.	System	WavLM-SV SIM	WavLM-SV Confusion [95% CI]	WavLM-SV ID	TitaNet-large SIM	TitaNet-large Confusion [95% CI]	TitaNet-large ID
En→Zh	Cascade, same data	0.82	24.0 [20.7, 27.0]	62.0	0.34	9.8 [5.8, 15.9]	85.9
	MIST	0.82	23.4 [19.2, 27.2]	62.2	0.35	10.2 [5.7, 16.3]	86.7
	MIST + LA2	0.83	19.8 [17.2, 22.0]	67.6	0.37	7.3 [5.4, 9.8]	90.5
	MIST-Simul	0.83	21.3 [19.6, 23.0]	66.3	0.38	6.7 [4.7, 8.8]	91.9
Zh→En	Cascade, same data	0.76	29.4 [24.3, 35.5]	55.9	0.20	24.6 [18.4, 30.4]	70.7
	MIST	0.76	30.1 [25.5, 35.0]	56.1	0.20	25.9 [20.2, 31.0]	68.8
	MIST + LA2	0.77	27.1 [23.5, 30.9]	60.9	0.20	23.3 [19.4, 27.8]	71.8
	MIST-Simul	0.77	27.9 [24.5, 31.6]	60.0	0.20	23.3 [19.7, 27.4]	71.4

Table 22: Voice of each output speaker's synthesised speech, matched to reference speakers per meeting by the target-language text (v5 matched Zh→En output against the Chinese source text, which made its Zh→En matching close to arbitrary; corrected here). SIM: cosine to the attributed speaker's own clean speech. Confusion: share of streams in multi-speaker windows closer to another speaker of the window (interval: meeting bootstrap). ID: the attributed speaker is the closest of all speakers of the meeting. Cosines of the two models are not comparable in scale.

System	cpASR↑	SIM↑	V-conf↓	Misr.↓	UTMOS↑	Abr.↓
En→Zh						
Prompted	10.13	0.79	39.6	14.5	–	–
Cascade (same data)	31.06	0.82	24.0	14.7	2.76	–
MIST (offline)	32.69	0.82	23.4	10.7	2.77	–
MIST + LA2	33.32	0.83	19.8	8.9	3.19	2.86
MIST-Simul	35.42	0.83	21.3	5.8	3.20	2.30
Zh→En						
Cascade (same data)	12.47	0.76	29.4	33.5	1.96	–
MIST (offline)	14.44	0.76	30.1	34.3	1.95	–
MIST + LA2	15.79	0.77	27.1	23.6	1.83	1.56
MIST-Simul	17.79	0.77	27.9	18.3	1.84	0.06

Table 23: Speech of the UniSS Talker: offline systems dubbed, simultaneous ones spoken block by block on one channel with causal voice prompts. cpASR: $\text{cpBLEU}_{\text{sess}}$ of the ASR transcripts; SIM: WavLM cosine to the attributed speaker (two speakers of a meeting: 0.75); V-conf: output speakers closer to another speaker of the window (%); Misr.: CER of the transcript against the Talker's input (%); Abr.: abrupt pauses per minute.

S Speaker Counts and Zero-Length Segments

On multi-speaker windows MIST under-counts and the cascade over-counts (Table 26): the cascade translates every diarizer turn, including two turns for one back-channel; MIST merges or drops a back-channel of at most 0.5 s in 41.3% of the windows whose least-heard speaker talks that briefly. Zero-length segments come mostly from such short responses and from the 20 s end of the time vocabulary (Table 27); enforcing a positive length after decoding leaves cpBLEU unchanged (it does not use time) and moves DER by at most 0.02 points. We did not train or decode with the constraint.

T Quality of the AliMeeting References

The English AliMeeting references are machine translations (Hy-MT2-30B) of the human transcripts; no human check was made (by decision, Limitations). We scored every reference utterance with GEMBA-DA (Kocmi and Federmann, 2023), the original Qwen3-Omni-30B-A3B Thinker prompted for a 0–100 score of the (source, translation) pair; a CometKiwi model was not available offline. Over 9,730 utterances the median score is 95; 5.0% score below 70 and 1.7% below 50. Leaving out every window that holds one of the lowest-scored 10% of references changes MIST − cascade from −0.35 to −0.05 and MIST-Simul − LocalAgreement from 0.87 to −0.58 $\text{cpBLEU}_{\text{sess}}$. An LLM judge is a weak substitute for a bilingual check.

U Relation to Prior Work

V v8: Overlap, Latency and Pre-registered Tests

Pre-registration. prereg_v8 was committed before any v8 test decode started; it fixes the systems, the overlap metrics and their thresholds, the causal relabelling rule, the twelve confirmatory tests and the wording for each outcome. Amendment 1, before any C2 or scheduling result, fixed the short-turn fallback and the playback policies. The speaker-blind Simul model is trained like MIST-Simul (same deadline-target rows, seed, 3,000 steps, batch 64), from the offline speaker-blind model with the speaker tokens zeroed. Table 30 lists the tests.

		20 s	coverage		silence-closed			close lat. (s)		attribution	re-entry	commit lag (s)
Set	System	windows	silence	20 s cut	≤2 s bound.	inside	repeat	silence	20 s	per 5 min	(≥60 s)	nested / other
En→Zh	MIST-Simul	15.3	75.2	68.5	75.0	75.6	7.8	2.01	4.94	87.5–92.6	85.4	1.36 / 1.24
	MIST + LA2	15.3	66.0	58.9	66.4	65.3	6.8	2.01	4.94	84.9–91.7	77.1	4.94 / 1.48
Zh→En	MIST-Simul	5.9	78.3	69.2	79.9	76.0	0.7	1.97	2.76	76.0–89.5	78.1	1.40 / 1.20
	MIST + LA2	5.9	75.8	65.2	77.2	73.8	0.5	1.97	2.76	83.9–89.8	74.9	3.86 / 1.48
AliMeeting	MIST-Simul	43.9	78.8	68.2	81.0	76.9	2.0	1.89	2.84	91.9–96.3	89.8	1.52 / 1.44
	MIST + LA2	43.9	65.1	52.9	67.2	63.2	1.8	1.89	2.84	88.8–94.2	84.5	6.50 / 2.92

Table 24: Whole-meeting online runs, by window type (%, except lags). 20 s windows: windows cut at the 20 s cap. Coverage: scored utterances that get an output segment, for utterances in silence-closed windows and in 20 s-cut windows, and (silence-closed) within 2 s of a window boundary or inside; repeat: covered utterances translated in two windows (the 0.2 s / 0.3 s padding). Close lat.: from the window's last reference speech to the moment its end is known (median). Attribution: share of covered utterances whose segment carries the right speaker after the meeting matching, lowest–highest over 5-minute bins with at least 50 utterances; re-entry: utterances of a speaker silent for at least 60 s. Commit lag (median, s after the segment's end) of segments nested in another speaker's segment vs. the rest (head-of-line blocking of the start-ordered output).

		hours		window	speech	overlapped		≤1 s utts	
Set	Mtgs	meetings	speech	audio h	kept	h	kept	h	kept
En→Zh	16	8.9	8.5	3.9	45.6	4.9	35.6	0.6	51.2
Zh→En	16	8.9	6.0	5.8	88.0	2.9	80.1	0.6	86.2
AliMeeting	20	10.7	12.2	7.5	63.0	8.0	45.0	0.5	65.1

Table 25: Coverage of the reference-window test sets: meeting time (to the last window), annotated speech, reference-window audio, and the share of speech, overlapped speech and utterances of at most 1 s that the windows keep (%). The whole-meeting online runs (Table 29) cover all of it.

All systems. Table 31 gives every system on reference windows and whole meetings with the overlap split, concurrent-speaker recall and collapse, recall of overlapped utterances under 0.5 s and units translated twice. MIST-Simul translates a unit twice more often than the offline model (8.1 vs. 1.4% in En→Zh), mostly when padding puts the same words into two consecutive reads' segments. Splitting a segment that spans two reference units in proportion to time (the v8 default) changes tcpBLEU by at most a few hundredths of a point against v7's rule.

Labels or translation? Relabelling MIST's own output with the diarizer, keeping its text, separates the two (re-review M09): on overlapped units its own labels score 34.56 against 31.94 with diarizer labels (Zh→En 19.72 vs. 16.74), while with diarizer labels on both sides its text leads the speaker-blind model's by 31.94 vs. 30.98. Most of the overlap gain is attribution, the rest translation.

Short-turn fallback (C2). For every window, a cascade segment of at most 1.5 s is added to MIST's output when no MIST segment of the same diarizer slot overlaps it within 0.5 s and no nearby MIST text has chrF ≥50 with it. On offline MIST this raises tcpBLEU in all three sets (Table 31) and concurrent-speaker recall to 86.2 / 84.2 / 86.7%; on MIST-Simul it raises recall but lowers tcpBLEU slightly, so by the pre-registered criterion it is adopted only for the offline model.

Latency. Table 4 queues the reads of a meeting on one decoder stream at the median single-stream read time measured in v6 (batch 1, two H200 cards, HuggingFace eager decoding). The wall-clock replay (B6) could only run beside another training job that held about 79 GiB and most of the compute of every card (prereg_v8 amendments 3–4), with the Thinker spread over three cards at batch 1, so its real-time factor is an upper bound on the single-stream one. On the one meeting completed (AliMeeting R8002_M8003, MIST-Simul) the median read took 3200 ms, 61% of the reads exceeded the 1 s cadence, and the measured real-time factor is 4.72. By the pre-registered rule this is inconclusive, not a negative result: contention only slows reads down, and an upper bound above 1 says nothing about the uncontended case. The user then stopped the remaining B6 runs, since further contended runs could not change that conclusion; two runs stopped partway (En→Zh MIST-Simul, 65 of 209 windows; AliMeeting LocalAgreement, 48 of 133) are not used. The real-time factors of Table 4 remain estimates from the measured single-stream read time, not a replay. LocalAgreement's reads take longer than the 1 s read period, so queued it falls behind without bound (per-speaker median 847.5 s). OpenSTBench-style start offset and average de-

Set	System	Wins	Exact	Under	Over	Wrong	under-count rate by the least-heard speaker ≤0.5 s	0.5–1 s	1–2 s	>2 s	overl.	not overl.
En→Zh	Cascade, same-data translator	831	78.7	14.9	6.4	2.4	28.3	12.1	6.0	5.1	14.8	22.2
	MIST (offline)	831	72.8	25.9	1.3	2.3	41.3	22.1	19.1	10.9	25.6	55.6
	MIST-Simul	831	77.0	21.1	1.9	6.5	36.0	17.9	14.1	6.4	20.7	55.6
	MIST (offline) + LA2	831	68.1	29.6	2.3	4.0	48.2	30.5	16.6	10.9	29.3	55.6
Zh→En	Cascade, same-data translator	1211	74.2	11.5	14.3	7.0	23.2	9.9	5.5	3.9	11.2	14.8
	MIST (offline)	1211	77.7	18.2	4.1	8.4	29.8	20.1	11.0	7.4	18.4	14.8
	MIST-Simul	1211	82.3	14.5	3.1	9.9	24.9	17.3	6.2	6.1	14.2	19.3
	MIST (offline) + LA2	1211	75.6	19.8	4.6	8.0	37.0	21.7	10.1	4.3	19.1	28.4
AliMeeting	Cascade, same-data translator	1418	88.0	5.1	6.9	0.6	21.6	3.9	1.3	1.5	5.1	0.0
	MIST (offline)	1418	83.2	14.1	2.7	1.0	36.7	16.3	8.5	6.8	14.1	16.7
	MIST-Simul	1418	83.6	12.8	3.6	3.2	36.2	16.6	8.3	3.2	12.9	8.3
	MIST (offline) + LA2	1418	80.3	17.5	2.2	2.5	44.5	25.0	10.3	5.7	17.5	16.7

Table 26: Speaker counts on multi-speaker reference windows (%). Under / Over: fewer / more output speakers than reference speakers; Wrong: right count, but a reference speaker gets no text after the session matching. The under-count rate is split by how long the window's least-heard speaker talks and whether that speaker overlaps another.

Set	System	$\alpha{=}\beta$ at 20 s	≤1 s ref.	cpBLEU before / after	DER before / after
En→Zh	MIST (offline)	2.6	2.3	13.7	42.81 / 42.81	13.99 / 13.98
En→Zh	MIST-Simul	4.3	0.7	6.8	42.63 / 42.63	11.04 / 11.02
En→Zh	Cascade, same-data translator	0.0	0.0	0.0	36.29 / 36.29	25.80 / 25.80
Zh→En	MIST (offline)	2.1	3.9	22.5	25.86 / 25.86	16.02 / 16.01
Zh→En	MIST-Simul	5.0	12.1	3.4	24.30 / 24.30	12.78 / 12.78
Zh→En	Cascade, same-data translator	0.0	0.0	0.0	22.36 / 22.36	15.50 / 15.50
AliMeeting	MIST (offline)	2.0	0.6	2.8	14.90 / 14.90	14.73 / 14.73
AliMeeting	MIST-Simul	7.3	8.8	1.3	12.97 / 12.97	10.81 / 10.80
AliMeeting	Cascade, same-data translator	0.0	0.0	0.0	15.25 / 15.25	13.10 / 13.10

Table 27: Zero-length output segments (start token = end token, % of segments), the share of them at the 20 s end of the time vocabulary and the share whose best-overlapping reference utterance lasts at most 1 s (% of zero-length segments), and the scores before / after enforcing $\beta \geq \alpha + 1$ (end moved one 80 ms bin later, or start one earlier at the window end).

lay of MIST-Simul (per-speaker channels, units spread uniformly over their block): 4.89 and 9.14 s in En→Zh; these are not comparable with numbers reported on single-speaker clips. Streams of more than 400 blocks (runaway repetition) were not synthesised: 1 on AliMeeting, none elsewhere; counted as never heard, they leave the percentiles unchanged at the reported precision.

Fixed lag and seeds. On the 8 AMI meetings of the v7 subset (prereg_v8 amendment 2), fixed lag 2 s on the prefix-trained model scores 40.28 tcpBLEU against MIST-Simul's 38.84 and LocalAgreement's 36.48 (En→Zh; Zh→En 24.69 / 24.91 / 23.88), but on overlapped units 32.24 vs. 35.18 (19.43 vs. 20.19), with a median text lag on overlapped units of 2.56 vs. 1.59 s. On the 10 AliMeeting subset meetings (completed after the other runs, prereg_v8 amendment 3, exploratory) fixed lag is ahead of MIST-Simul overall and on overlapped units (12.83 / 10.17 vs. 11.55 / 8.97; LocalAgreement 10.92 / 6.24), again at a longer lag on overlapped units (2.48 vs. 1.45 s). Whole-meeting tcpBLEU of MIST-Simul over training seeds: 40.14 ± 0.52 (En→Zh, 3 seeds; overlapped 35.32 ± 0.42), 24.38 ± 0.22 (Zh→En, 2), 11.03 ± 0.63 (AliMeeting, 2).

Playback scheduling. Table 32 replays MIST-Simul's committed blocks under four single-channel policies and the per-speaker layout. Captions for short overlapped interjections and skipping stale blocks bound the lag of what is spoken, but on AliMeeting, where speakers overlap most, only 35% of the units are then spoken.

Voice prompts (B8). On the 8 whole AMI En→Zh meetings of v7's second half (3,318 output streams), the Talker was run again with only the most recent 1 s of a slot heard alone as its prompt instead of up to 8 s, nothing else changed. The short prompt makes the voices less like the attributed speaker (WavLM SIM 0.79 vs. 0.83, meeting-bootstrap intervals [0.76, 0.82] and [0.81, 0.86]), more often closer to another speaker of the window (30.6 vs. 20.0%) and slightly less intelligible (ASR-BLEU against the Talker's input 67.53 vs. 68.96), so we keep 8 s. At the first commit 98.9% of the streams have 1 s heard alone and 95.1% a full 8 s. Holding a slot's speech until 1 s of it has been heard alone, instead of starting with the fallback prompt, would delay 1.0% of the streams (0.36% of the blocks), but those by a median 20.3 s (90th percentile 235 s): slots that rarely talk alone. Speaking at once is the better default.

Function	Prior work	What this paper adds
Concurrent speakers	DiariST (Yang et al., 2024) (t-SOT serialisation, text only, no timestamps); STAC-ST (Zuluaga-Gomez et al., 2023) (two-party, offline)	every speaker of an overlapping window written with speaker, time and translation; overlap-aware scores show the gain over one-speaker-per-span relabelling (§5.2)
Speaker tokens interleaved with acoustic tokens	G-STAR (Peng et al., 2026): multi-talker ASR with a persistent speaker stream	the same interleaving for translation; its benefit is measurable on overlapped speech, not on single-speaker speech
Simultaneous policy	LocalAgreement (Liu et al., 2020; Polák et al., 2022), fixed lag; SimulS2S-LLM (Deng et al., 2025)	a wait token trained from source-timed chunk deadlines; with queued reads it keeps pace with the audio where LA2 does not; quality vs. LA2 on the same model in Table 3
Voice in the output	X-Translator (Zhao et al., 2026) (one speaker per recognised span), single-speaker SimulS2ST (Seamless Communication et al., 2023; Labiausse et al., 2025; Cheng et al., 2025a); user study (Ma et al., 2026)	causal per-speaker voice prompts from the diarizer's solo regions; one vs. per-speaker channels; no listening test
Meeting evaluation	cpWER / tcpWER (Watanabe et al., 2020; von Neumann et al., 2023); overlap decomposition of tcpWER; unsegmented evaluation (Papi et al., 2025)	cpBLEU, tcpBLEU with overlap / single-speaker split and concurrent-speaker recall; whole-meeting runs with queued reads and listener-side lag

Table 28: What each part of MIST takes from prior work and what this paper adds. The comparison with other systems is by capability; we do not re-run them under our protocol, and no cross-paper numerical comparison is made.

			tcpBLEU					text end lag (s)		listener lag, completion (s)		
Set	System	Mtgs	windows	online	speech	Omit	Ins.	p50	p90	1 ch. p50	per-spk p50	per-spk p90
En→Zh	MIST-Simul	16	42.45	40.09	29.07	20.6	4.9	1.87	2.82	141.8	6.5	11.2
	MIST + LA2	16	31.41	34.26	27.49	25.7	7.0	2.30	6.09	255.1	11.0	104.7
Zh→En	MIST-Simul	16	23.96	24.53	16.60	18.6	6.3	1.94	2.41	26.0	7.9	14.9
	MIST + LA2	16	24.30	23.94	14.90	21.0	7.8	2.01	5.57	27.5	7.4	15.8
AliMeeting	MIST-Simul	20	13.34	11.46	9.43	13.0	3.1	1.62	2.49	604.4	18.0	306.3
	MIST + LA2	20	12.41	11.09	9.14	15.8	3.6	2.77	7.91	734.9	36.7	194.3

Table 29: Whole meetings run online, all test meetings (AMI 16 per direction, AliMeeting 20; no reference boundary). tcpBLEU of the same system on the reference windows of the same meetings (windows), run online (text), and of the ASR of its synthesised speech (speech); Omit / Ins.: unscored reference units / inserted segments (%); text end lag: last commit of a reference unit minus its end, including the delay until a window's end is detected; listener lag: end of a block's speech minus the end of its source segment, with measured computation and synthesis, on one interpreter channel (1 ch.) or one channel per speaker (Appendix Q).

Threshold flips (B5). The causality check (Appendix K) found 8 distinct 80 ms frames whose 0.5 activity decision differs between the prefix run and the dump. Rebuilding the online windows with the prefix values changes all window boundaries (none moves). The 8 windows holding a flipped frame were decoded again, each together with a copy using the original posteriors in the same batch: 3/8 are token-identical; an unchanged copy decoded in the same batch is identical in 5/8. With so few windows the two cannot be told apart: the flips change outputs no more than re-decoding does, but this check cannot show that they change nothing.

Full-coverage references. The reference windows keep 45.6% of the En→Zh test speech. Every test utterance has a selected translation (Appendix A), so we also score the whole-meeting runs on all 6563 non-hesitation units: MIST-Simul 39.24, LocalAgreement on the offline model 27.01; overlapped units 33.56 vs. 16.61. These are still machine translations.

Test	Set	Comparison (whole meetings, online)	Δ	95% interval	p_{Holm}	
E-ov1-en2zh	En→Zh	tcpBLEU-ov, MIST-Simul − causal relabel	8.54	[3.82, 14.28]	0.000	holds
E-ov2-en2zh	En→Zh	concurrent-speaker recall, same pair	0.168	[0.1451, 0.1939]	0.000	holds
E-lat-en2zh	En→Zh	text lag, overlapped − single (<0.5 s)	-0.36	[-0.425, -0.299]	0.000	holds
E-pol-en2zh	En→Zh	text lag, LA2 (same model) − MIST-Simul	0.81	[0.553, 1.081]	0.000	holds
E-ov1-zh2en	Zh→En	tcpBLEU-ov, MIST-Simul − causal relabel	8.67	[2.63, 12.94]	0.002	holds
E-ov2-zh2en	Zh→En	concurrent-speaker recall, same pair	0.258	[0.2185, 0.3022]	0.000	holds
E-lat-zh2en	Zh→En	text lag, overlapped − single (<0.5 s)	-0.21	[-0.264, -0.167]	0.000	holds
E-pol-zh2en	Zh→En	text lag, LA2 (same model) − MIST-Simul	0.23	[0.107, 0.374]	0.000	holds
E-ov1-ali	AliMeeting	tcpBLEU-ov, MIST-Simul − causal relabel	0.63	[-2.43, 3.51]	0.381	tie / fails
E-ov2-ali	AliMeeting	concurrent-speaker recall, same pair	0.172	[0.1313, 0.2049]	0.000	holds
E-lat-ali	AliMeeting	text lag, overlapped − single (<0.5 s)	-0.23	[-0.325, -0.119]	0.000	holds
E-pol-ali	AliMeeting	text lag, LA2 (same model) − MIST-Simul	1.89	[1.327, 2.496]	0.000	holds

Table 30: The twelve confirmatory tests of prereg_v8, Holm-corrected together. A test holds when $p_{\text{Holm}} < 0.05$ and the interval excludes 0 (E-lat: lies below the 0.5 s bound). Exact sign-flip permutation over meetings; intervals from 10,000 meeting resamples, BLEU differences with a decoder-noise draw per resample.

System	tcpBLEU↑ all	tcpBLEU↑ overlap	tcpBLEU↑ single	concurrent speakers recall↑	concurrent speakers collapse↓	short ≤0.5 s↑	short dup.↓
En→Zh, reference windows							
Cascade, same data	36.13	26.21	41.72	89.3	9.1	67.8	1.6
Cascade, turn-trained	34.61	32.77	35.28	81.9	14.6	56.8	1.3
Speaker-blind + relabel (offline)	41.39	30.98	46.04	51.8	40.9	34.8	2.1
MIST (offline)	42.06	34.56	45.35	76.3	20.3	60.0	1.4
MIST text, diarizer relabelling	41.17	31.94	45.31	53.3	40.5	35.7	2.5
MIST (offline) + fallback	42.53	34.67	45.52	86.2	12.7	69.1	2.5
MIST-Simul model + LA2	38.96	27.38	45.04	75.9	22.5	51.7	3.6
MIST-Simul	42.45	36.00	44.71	83.8	15.5	56.9	8.1
En→Zh, whole meetings online							
MIST (offline) + LA2	34.30	22.34	41.86	73.4	23.8	50.7	1.4
MIST-Simul model + LA2	39.31	32.91	40.91	75.5	22.8	50.4	4.0
MIST-Simul model, lag 2 s†	40.28	32.24	43.65	74.4	24.0	47.8	6.2
Speaker-blind Simul + causal relabel	35.22	27.17	38.80	65.5	30.6	37.9	9.3
MIST-Simul	40.09	35.71	41.39	82.3	16.2	55.0	8.6
Zh→En, reference windows							
Cascade, same data	21.96	14.36	26.31	87.4	12.0	67.5	1.0
Cascade, turn-trained	23.20	19.39	24.76	83.8	15.4	64.1	1.1
Speaker-blind + relabel (offline)	24.36	16.90	27.63	50.6	44.8	34.9	5.5
MIST (offline)	25.43	19.72	27.80	79.0	19.1	62.2	4.2
MIST text, diarizer relabelling	24.46	16.74	27.86	46.6	48.6	30.9	5.4
MIST (offline) + fallback	25.74	20.07	28.08	84.2	15.0	67.2	4.7
MIST-Simul model + LA2	24.41	18.26	26.94	80.4	18.7	63.9	5.3
MIST-Simul	23.96	15.59	26.99	82.4	17.1	68.3	7.3
Zh→En, whole meetings online							
MIST (offline) + LA2	23.94	18.88	26.04	78.7	19.8	61.4	4.1
MIST-Simul model + LA2	23.96	17.89	26.43	79.4	19.9	65.2	5.1
MIST-Simul model, lag 2 s†	24.69	19.43	26.80	76.8	22.3	66.0	6.7
Speaker-blind Simul + causal relabel	19.33	9.06	26.72	54.4	42.4	35.7	7.5
MIST-Simul	24.53	17.73	26.68	80.2	19.0	68.1	7.0
AliMeeting, reference windows							
Cascade, same data	15.70	9.79	17.96	93.3	6.5	60.6	2.1
Cascade, turn-trained	13.27	12.03	13.68	89.2	10.1	52.9	1.6
Speaker-blind + relabel (offline)	16.27	12.04	18.26	58.0	35.2	20.9	6.2
MIST (offline)	15.19	12.10	16.60	79.2	15.0	46.3	4.4
MIST text, diarizer relabelling	14.91	11.34	16.59	55.9	36.3	19.6	5.9
MIST (offline) + fallback	15.44	12.64	16.65	86.7	11.8	59.4	5.3
MIST-Simul model + LA2	13.80	10.13	15.34	80.2	18.4	40.0	10.1
MIST-Simul	13.34	9.90	14.95	83.4	15.7	46.1	17.1
AliMeeting, whole meetings online							
MIST (offline) + LA2	11.11	7.19	13.39	75.6	20.1	40.2	5.3
MIST-Simul model + LA2	11.91	7.60	14.30	76.7	21.4	36.8	9.7
MIST-Simul model, lag 2 s†	12.83	10.17	14.00	84.8	13.9	41.2	17.2
Speaker-blind Simul + causal relabel	12.67	8.55	14.52	63.0	31.7	22.9	18.7
MIST-Simul	11.47	9.18	12.42	80.2	17.4	46.1	17.4

Table 31: Overlap-aware scores (§4). Overlap: reference units with ≥20% of their time overlapped by another speaker; single: the rest. Concurrent speakers: pairs of utterances of two speakers overlapping ≥0.5 s; recall: both translated and attributed; collapse: one side missing and covered by the other side's output speaker. Short: recall of overlapped utterances ≤0.5 s. Dup.: units translated twice (chrF ≥50 between two segments, %). † On the v7 meeting subset only (8 AMI / 10 AliMeeting meetings); compare with Appendix V, not with the other rows.

Playback	En→Zh p50	En→Zh p90	En→Zh spoken	Zh→En p50	Zh→En p90	Zh→En spoken	AliMeeting p50	AliMeeting p90	AliMeeting spoken
one channel, as is	142.4	772.6	100	26.4	313.0	100	604.6	2481.2	100
short interjections as captions	43.9	320.6	91	13.3	47.5	94	441.0	1630.3	93
skip blocks >10 s late	9.7	11.8	80	9.8	13.1	82	10.8	17.2	35
b + c	9.1	11.6	80	9.4	12.9	82	10.8	17.4	35
one channel per speaker, mixed	7.4	12.2	100	8.2	15.1	100	23.2	314.1	100

Table 32: Playback scheduling of MIST-Simul's whole-meeting output (reads queued; nothing re-synthesised, so the speech of every played block is unchanged): listener completion lag (s) of the units played, and the share of units spoken (%; the rest are shown as captions). prereg_v8 amendment 1.